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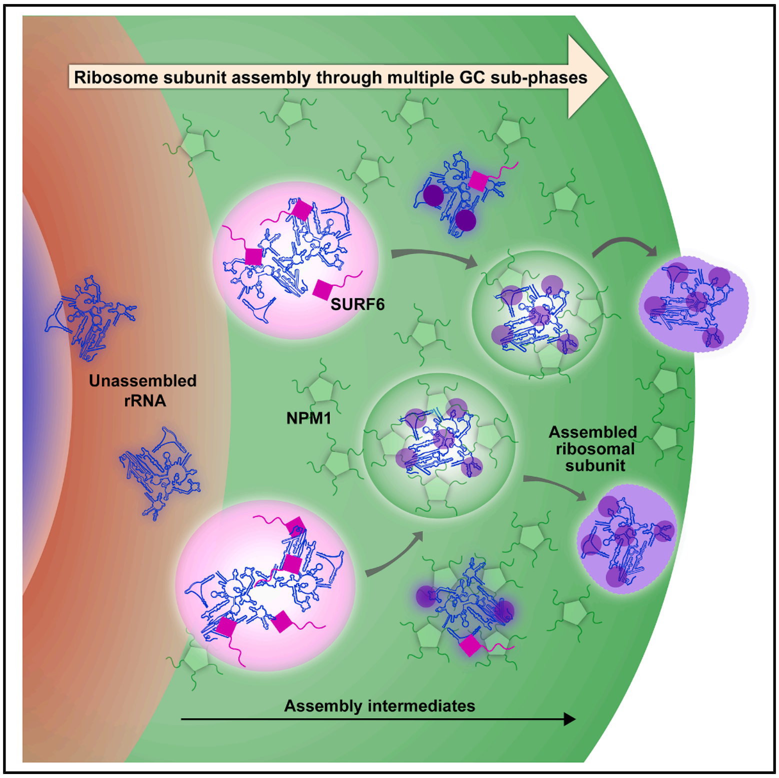
Granular component sub-phases direct ribosome biogenesis in the nucleolus

Graphical abstract



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In brief

Dogra et al. use super-resolution microscopy to uncover the molecular origins of spatial heterogeneity in the nucleolar granular component (GC). They show that differential interactions of nucleolar proteins, including ribosomal assembly factors, with ribosomal RNA generate distinct GC sub-phases that enable an assembly-line-like mechanism governing ribosome biogenesis.

Highlights

- Nucleolar GC heterogeneity arises through the multiphase behavior of biomolecules
- Differential affinities of NPM1 and assembly factors for rRNA form GC sub-phases
- *In vitro*-reconstituted GC sub-phases exhibit distinct physicochemical properties
- The multiphase nature of the GC orchestrates ribosome subunit assembly

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Granular component sub-phases direct ribosome biogenesis in the nucleolus

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SUMMARY

The hierarchical, multiphase organization of the nucleolus underlies ribosome biogenesis. Ribonucleoprotein particles that regulate ribosomal subunit assembly are heterogeneously distributed in the nucleolar granular component (GC). However, the molecular origins of the GC's spatial heterogeneity and their link to ribosome subunit assembly remain poorly understood. Here, using super-resolution microscopy in DLD-1 cells, we uncover that key GC biomolecules—NPM1, SURF6, and ribosomal RNA (rRNA)—are heterogeneously localized within GC sub-phases. *In vitro* reconstitution with *E. coli*- and human-derived rRNA revealed that these GC biomolecules form multiphase condensates with a SURF6/rRNA-rich core and NPM1-rich shell, providing a mechanistic basis for this heterogeneity. SURF6's association with rRNA weakens upon ribosome subunit assembly, enabling NPM1 to extract assembled subunits from condensates, suggesting an assembly-line-like mechanism of subunit efflux from the GC. Our results establish a framework for understanding the GC's heterogeneous structure and reveal how its distinct sub-phases facilitate ribosome subunit assembly.

INTRODUCTION

The discovery of phase-separated biomolecular condensates changed our understanding of how biomolecules are organized within cells to perform their biological functions.[1–4] Biomolecular condensates are dynamic, membraneless assemblies that form through weak, transient multivalent interactions between proteins, nucleic acids, and other biomolecules.[4–6] One example, the nucleolus, coordinates ribosome biogenesis through its hierarchical spatial organization that supports the assembly of pre-ribosomal particles.[7–12] Human nucleoli display three liquid-like sub-compartments, termed the fibrillar center (FC), dense fibrillar component (DFC), and granular component (GC),[10] that remain spatially separate due to differences in their material properties, particularly surface tension.[13] Within the FC during interphase, RNA polymerase I and associated transcription factors initiate transcription of tandemly repeated ribosomal RNA (rRNA) genes (ribosomal DNA or rDNA).[14] Nascent pre-rRNAs undergo extensive posttranscriptional modifications and processing within the DFC.[14] During this stage, more than 200 assembly factors facilitate cleavage, modification, and processing of pre-rRNA to yield mature rRNAs,[7,10,15] which move into the GC and assemble with ribosomal proteins to form ribosomal subunits.[16] The separation of the FC, DFC, and GC phases within the nucleolus enables the compartmentalization of the different processes underlying ribosome biogenesis.

The liquid-like nature of the different nucleolar sub-compartments enables ribosomal components and supporting biomolecules to move within and between them during ribosome subunit assembly.[13,17,18] Intrinsically disordered regions (IDRs) in numerous nucleolar proteins, including nucleophosmin (NPM1), nucleolin (NCL), and fibrillarin (FBL), interact with other nucleolar components, such as rRNA, small nucleolar RNAs (snoRNAs), and/or ribosomal proteins, and undergo phase separation to establish the distinct sub-compartments.[13,17,19–22] rRNA processing occurs in distinct spatial layers within the nucleolus, and specific defects in rRNA processing can disrupt this multi-layered organization.[23] Various factors, such as concentration, posttranslational modifications, and cellular state, influence the phase separation properties of nucleolar proteins.

Changes in these factors can modulate biomolecular interactions and phase separation, leading to the formation or dissolution of condensed phases within the nucleolus.[8] However, our understanding of the internal structure and dynamics of the individual sub-compartments of the nucleolus, e.g., the GC, through the lens of phase separation remains incomplete.

Recent advances in super-resolution microscopy have created opportunities for investigating the substructural organization of nucleolar compartments.[24] However, how the structural heterogeneity in the GC contributes to the multiple steps of ribosome subunit assembly[15] remains unexplained. For example, the GC is so named because it appears to comprise disorganized, heterogeneous granules in electron microscopy images.[25,26] These granules are thought to contain rRNA, ribosomal proteins, and protein factors involved in ribosome subunit assembly.[12,27,28] Given the roles of differential biomolecular composition and material properties in the hierarchical organization of the primary nucleolar sub-compartments, we hypothesized that the spatial heterogeneity of the GC arises from sub-phases composed of distinct subsets of its biomolecular components. To test this hypothesis, we examined three key GC components: rRNA, which forms the catalytic core of ribosomes and serves as a structural scaffold for ribosomal proteins; NPM1, the highly abundant multivalent scaffold protein[20]; and surfeit locus protein 6 (SURF6), an essential ribosomal subunit assembly factor.[15] NPM1 establishes the liquid-like environment of the GC through phase separation, while SURF6 is an arginine (Arg)- and lysine (Lys)-rich nucleolar protein that prevents premature folding of key rRNA elements during pre-60S biogenesis.[15] Although NPM1 phase separates individually with rRNA and SURF6,[17,20,21] the behavior of more complex mixtures was unknown, whereas a truncated form of SURF6 (SURF6-N) was previously shown to form homogeneous ternary condensates with NPM1 and rRNA.[18] We probed these components in the GC using structured illumination microscopy (SIM), which revealed preferential colocalization of rRNA with SURF6 and partial spatial delocalization of rRNA from NPM1. Although the role of phase separation in the formation of the GC was previously established,[17,18,20] the origins of its spatial heterogeneity and contribution to ribosome subunit assembly remain elusive. Here, we developed an *in vitro* model system that recapitulates the GC sub-phases observed in cells. In this system, NPM1, SURF6, and rRNA form condensates with two coexisting dense phases that reflect the spatial heterogeneity observed in the GC. These so-called multiphase condensates exhibit a core-shell architecture, with rRNA and SURF6 enriched in a hydrophobic core and NPM1 enriched in a surrounding hydrophilic shell phase. Both phases retain liquid-like material properties and dynamically adapt to component changes, both *in vitro* and in cellular nucleoli. Importantly, we observed that the packaging of rRNA with ribosomal proteins weakens interactions with SURF6, enabling NPM1 to extract assembled rRNA from the core condensate phase. Our results indicate that, as rRNA assembles with ribosomal proteins, it flows between at least two GC sub-phases enriched in SURF6 and NPM1, respectively, en route to the nucleoplasm. Sequence analyses, as well as supporting cellular and biochemical data for SSF1, indicate that many additional assembly factors can interact with rRNA similarly to SURF6, creating a GC sub-phase for early ribosomal subunit assembly steps. These findings reveal how spatial heterogeneity, arising from the complex interplay of molecular interactions within the GC, facilitates ribosome subunit assembly.

RESULTS

Super-resolution imaging reveals the molecular underpinnings of spatial heterogeneity in the nucleolar GC

Although the nucleolar GC has long been known to be structurally heterogeneous, comprising ribonucleoprotein (RNP) complexes,[9,10,25,26] the physical origins of its granularity have remained elusive. The length scale of the granular structure of the GC is below the traditional light microscopy diffraction limit. Therefore, we used super-resolution SIM fluorescence imaging[29,30] to resolve nucleoli of human DLD-1 colorectal adenocarcinoma cells engineered using CRISPR-Cas9 to express endogenous NPM1 tagged with monomeric, enhanced green fluorescent protein (mEGFP) and FBL tagged with monomeric red fluorescent protein (mCherry; DLD-1$^{\text{NPM1-G/FBL-R}}$ cells).[31] We confirmed the hierarchical structure of nucleoli in these cells using NPM1 as a GC marker, FBL for the DFC, and upstream binding factor (UBF) for the FC (Figures 1A and S1A; see STAR Methods), with the spatial heterogeneity of NPM1 evident within the GC.

We next investigated the molecular basis of heterogeneity in the GC by imaging NPM1, SURF6, and rRNA using SIM. NPM1 is a GC scaffold protein that undergoes phase separation with ribosomal and non-ribosomal proteins, including SURF6, and facilitates ribosome subunit assembly and efflux from the nucleolus.[17,18,20,21] NPM1 contains a pentameric N-terminal oligomerization domain (OD), a central IDR with alternating acidic and basic tracts, and a C-terminal nucleic acid-binding domain (NBD) (Figures 1B, top and S1B). Acidic tracts in the NPM1 IDR mediate binding to Arg-rich basic tracts in other proteins, including SURF6, and the NBD mediates binding to rRNA (Figure S1B).[17,20,21] SURF6 is an essential Arg- and Lys-rich disordered protein displaying multiple basic tracts (Figures 1B, bottom and S1B) that functions as an early assembly factor in the maturation of pre-60S ribosomal subunits.[15] Although FBL, NPM1, and SURF6 were visualized using immunofluorescence (see STAR Methods), newly transcribed rRNA was visualized using metabolic labeling with 5-ethynyluridine (EU) through a 30-min pulse, followed by click chemistry detection with an Alexa-647 fluorophore.[32] SIM imaging revealed the structural heterogeneity of the nucleolus (Figures 1C and S1C), with the fluorescence intensity profile for a vector extending from one DFC (marked by FBL) through the GC to the nucleolar periphery revealing heterogeneous localization of NPM1, SURF6, and rRNA in the GC (Figure 1D). This structural heterogeneity was not resolved in our prior studies that used conventional confocal fluorescence microscopy.[18] Because SURF6 is an early assembly factor in pre-60S biogenesis, which begins at the interface of the DFC and GC, we performed this analysis within segmented DFC regions and the immediately surrounding GC regions (Figure S1D). Manders' overlap coefficient analysis demonstrated that rRNA is significantly more colocalized with SURF6 than with NPM1 (Figures 1E, S1E, and S1F), whereas SURF6 and NPM1 are highly colocalized with each other (Figure S1G). To further explore the

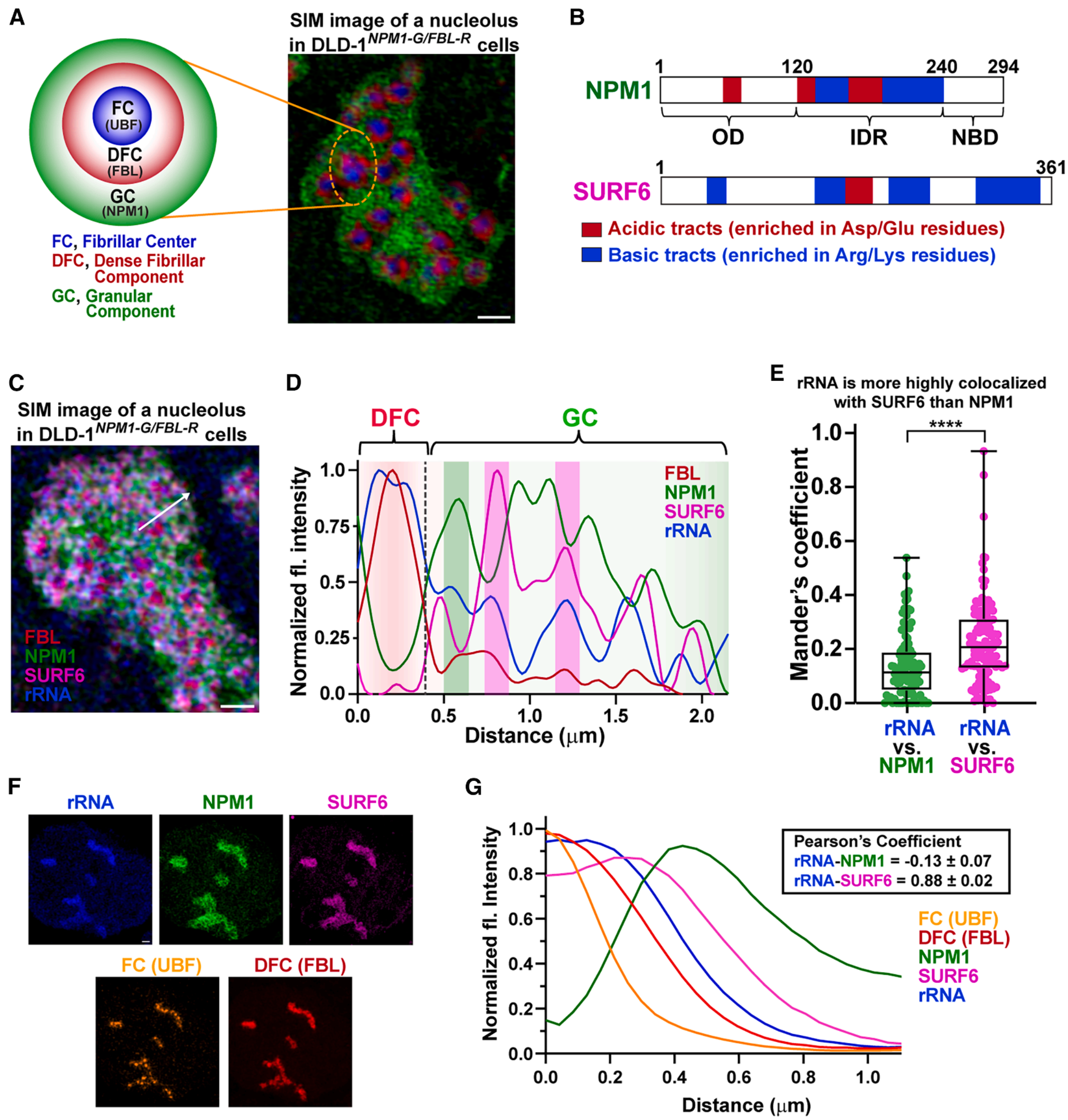

Figure 1. Super-resolution imaging resolves the molecular basis for spatial heterogeneity in the GC

(A) Schematic of the three-layered nucleolar structure (left). The innermost layer is the fibrillar center, surrounded by the dense fibrillar component within the granular component. 2D SIM image of DLD-1$^{NPM1\text{-}G/FBL\text{-}R}$ cells immunostained to probe NPM1 (green) and FBL (red) with anti-GFP and anti-RFP nanobodies conjugated to AF488 and AF568, respectively, and UBF (blue, pseudocolored) with anti-UBF-AF647 antibody; scale bar, 1 μm.

(B) Domain architecture of NPM1 and SURF6. NPM1 contains an oligomerization domain (OD) with an acidic tract (red), an intrinsically disordered region (IDR) comprising two acidic tracts interspersed with basic tracts (blue), and a nucleotide-binding domain (NBD). SURF6 is N-terminally disordered with a helical C terminus and exhibits four Arg-rich basic tracts and a single acidic tract.

(C) SIM image of a DLD-1$^{NPM1\text{-}G/FBL\text{-}R}$ cell nucleolus immunostained for SURF6, NPM1-mEGFP, mCherry-FBL, and EU-labeled rRNA (pulse EU-labeling for 30 min, without chase [t = 0 min]) labeled with AF647; scale bar, 1 μm. See STAR Methods.

(D) Normalized fluorescence intensity line plot (white arrow in C) for FBL, NPM1, SURF6, and rRNA. Green bar: NPM1/rRNA-enriched region; magenta bars: SURF6/rRNA-enriched regions.

(E) Manders' overlap coefficients for rRNA vs. NPM1 and rRNA vs. SURF6 in segmented DFC/inner-GC regions, n = 112 DFC/GC regions, six nucleoli; box plot shows the median (center line), 25th–75th percentiles (box), and minimum–maximum values (whiskers) ;the two-tailed paired t test was applied (****, $p < 0.0001$).

(F) Super-resolution Airyscan image of a DLD-1$^{NPM1\text{-}G/FBL\text{-}R}$ cell nucleolus immunostained for SURF6, UBF, NPM1-mEGFP, mCherry-FBL, and EU-rRNA (pulse EU-labeling for 30 min, without chase [t = 0 min]), labeled with AF647; scale bar, 1 μm. See STAR Methods.

(G) Min-max-normalized intensities for UBF (FC), FBL (DFC), SURF6, NPM1, and EU-rRNA (pulse EU-labeling for 30 min, without chase [t = 0 min]) by distance from the FC center. Inset: Pearson's coefficients for rRNA vs. NPM1 and rRNA vs. SURF6. Values represent mean ± SEM, n = 41 cells.

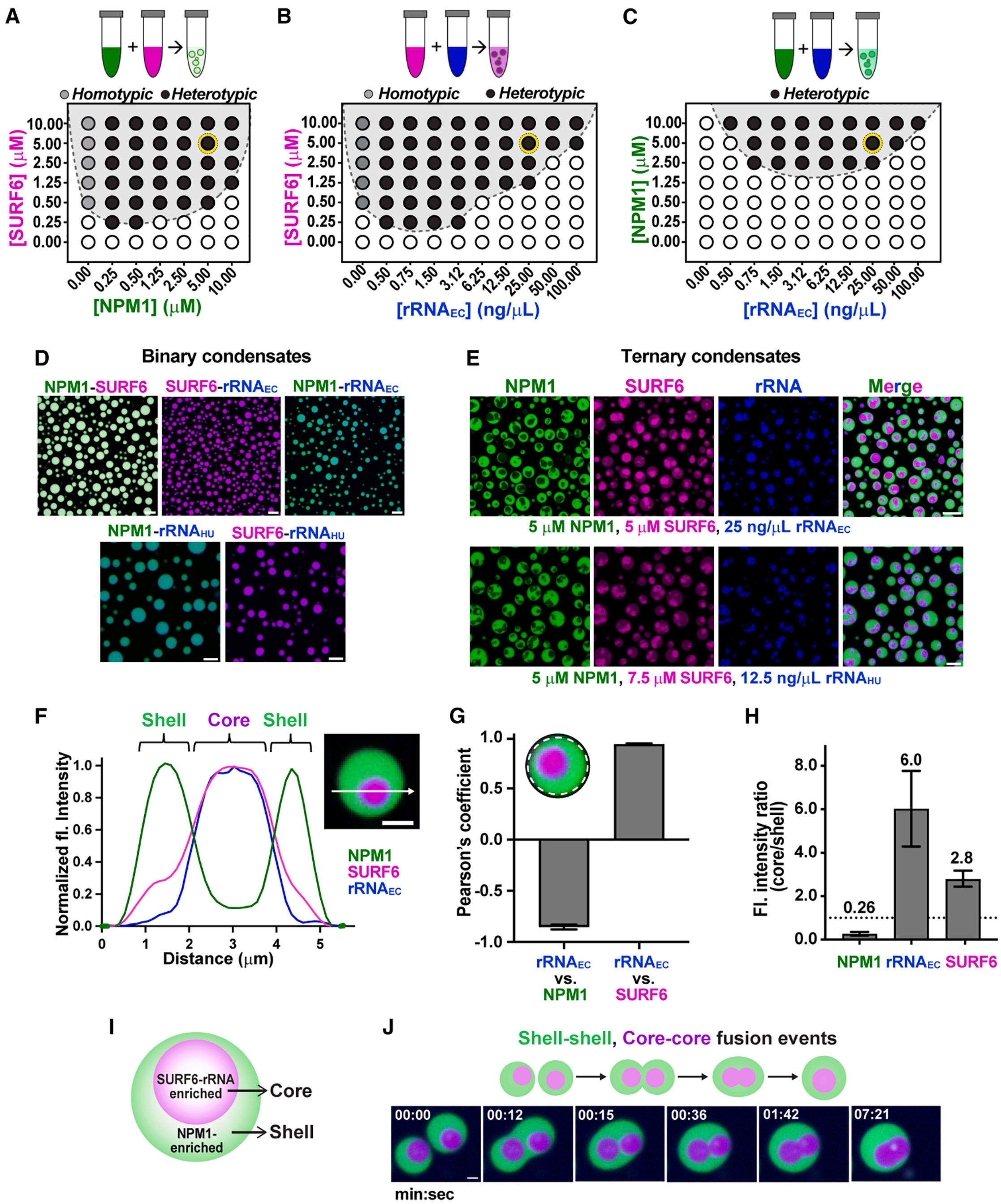

Figure 2. Competing interactions between NPM1, SURF6, and rRNA drive formation of multiphase condensates

(A–C) LLPS phase diagrams for (A) NPM1-SURF6, (B) SURF6-rRNA$_{EC}$, and (C) NPM1-rRNA$_{EC}$ mixtures in TBS-150. Shaded gray: condensate-forming conditions using confocal fluorescence microscopy (see Figure S2 for images). Light and dark gray circles: homotypic and heterotypic phase separation, respectively; open circles: no phase separation.

(D) Representative images of merged channels for binary condensates formed for NPM1-SURF6, SURF6-rRNA$_{EC}$ (rRNA$_{EC}$: total *E. coli* rRNA), and NPM1-rRNA$_{EC}$ at concentrations indicated by the yellow circles in panels A–C, respectively; scale bar, 10 μm. NPM1-rRNA$_{HU}$ (rRNA$_{HU}$: total human rRNA) and SURF6-rRNA$_{HU}$ at 5 μM NPM1, 7.5 μM SURF6, and 12.5 ng/μL rRNA$_{HU}$; scale bar, 10 μm. NPM1 was conjugated with AF488 (green) and SURF6 with AF647 (magenta). rRNA$_{EC/HU}$ was stained with SYTO 40 (blue).

(E) Fluorescence micrographs of ternary multiphase condensates of NPM1-SURF6-rRNA$_{EC/HU}$ (EC: top, HU: bottom); scale bar, 10 μm.

(F) Normalized fluorescence intensity line plot across the image of multiphase condensate (white arrow) for NPM1, SURF6, and rRNA$_{EC}$; scale bar, 2 μm. The shell is enriched in NPM1, and the core is enriched in SURF6 and rRNA$_{EC}$.

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spatial distribution of these components, we performed super-resolution Airyscan imaging and radial distribution function (RDF) analyses.[23,33] We labeled UBF to identify FC centers and determined the RDF profiles for FBL, rRNA-EU, SURF6, and NPM1 (Figures 1F, 1G, and S1H). Importantly, Pearson's correlation coefficient (PCC) values obtained from the RDF traces for rRNA and NPM1 vs. rRNA and SURF6 corroborated our SIM imaging observations (Figures 1G and S1H), establishing a baseline for characterizing the flux of rRNA through nucleolar subcompartments. These results indicate that differential localization of NPM1, SURF6, and rRNA underlies the structural heterogeneity of the nucleolar GC. Next, we investigated the mechanistic origins of this heterogeneity using *in vitro* reconstitution with the three GC biomolecules.

Competing interactions between NPM1, SURF6, and rRNA create spatial heterogeneity within multiphase condensates

NPM1, SURF6, and rRNA coexist within the GC through phase separation mediated by different multivalent interactions. We previously showed that acidic tracts in the NPM1 IDR interact with Arg-rich motifs in other proteins, including SURF6 (Figure 1B), and the NBD binds total *E. coli* rRNA ($rRNA_{EC}$).[17,20,21] We also showed that the Arg- and Lys-rich N-terminal IDR of SURF6 interacts with $rRNA_{EC}$ and the acidic tracts of NPM1.[17] To quantify pairwise interactions of these components, we generated binary phase diagrams using confocal fluorescence microscopy (Figures 2A–2C and S2A–S2C). NPM1 and SURF6 were visualized through conjugation with Alexa Fluor 488 (NPM1-AF488) and Alexa Fluor 647 (SURF6-AF647), respectively, and $rRNA_{EC}$ was visualized using SYTO 40 dye (Figure S2D). SURF6 exhibited homotypic phase separation and heterotypic phase separation with NPM1 and $rRNA_{EC}$ and formed heterotypic condensates with $rRNA_{EC}$ at a 10-fold lower concentration than did NPM1 (Figures 2A–2D and S2A–S2D). Similarly, total human rRNA ($rRNA_{HU}$) yielded comparable results (Figures 2D and S2E). Next, we used the concentrations marked by yellow circles in the phase diagrams (Figures 2A–2C) to test the behavior of a ternary mixture of the components. Strikingly, we observed the formation of multiphase condensates that exhibit a core-shell architecture (Figure 2E, top), which were recapitulated with $rRNA_{HU}$ at a slightly different concentration (Figure 2E, bottom). Fluorescence intensity plots showed that rRNA is colocalized with SURF6 in the core and not with NPM1 in the shell (Figure 2F). PCC analysis revealed strong $rRNA_{EC}$ correlation with SURF6 and anti-correlation with NPM1 (Figure 2G), with $rRNA_{EC}$ and SURF6 6-fold and ~3-fold enriched in the core, respectively, and NPM1 ~4-fold enriched in the shell (Figures 2H and 2I). We next performed time-lapse imaging and observed that shells fuse within seconds, whereas cores coalesce within minutes (Figure 2J). The initially dumbbell-like fused shells and cores adopted a near-spherical shape over time (Figure 2J; Video S1). Together, our results show that SURF6 mediates the formation of multiphase condensates through phase separation involving homotypic interactions and differential heterotypic interactions with rRNA and NPM1. To understand the driving forces underlying their core-shell architecture, we next investigated the physicochemical properties of the two phases of reconstituted multiphase condensates.

Disparate physicochemical properties underlie the core-shell architecture of multiphase condensates

Nucleolar sub-compartments have different material and physicochemical properties due to their different biomolecular compositions.[13] These properties, including surface tension, diffusivity, and viscoelasticity,[13,33] influence the dynamic exchange of components within the nucleolus. We hypothesized that intrinsic properties of the SURF6-rRNA core phase underlie the core-shell condensate architecture seen when NPM1 is also present. To test this, we separately formed core (SURF6-$rRNA_{EC}$) and shell (NPM1-SURF6) mimics and combined them (Figure 3A). Interestingly, within minutes, the cores were engulfed by the shell phase (Figure 3B; Video S2). Next, we individually monitored the accumulation of supplemental SURF6 or $rRNA_{EC}$ in the cores within pre-made multiphase condensates. To examine SURF6 accumulation, we initially formed condensates with unlabeled SURF6 and then, after 2 h, added partly fluorescently labeled SURF6 (fl-SURF6, 2% of the total SURF6; Figure 3C). The supplemental SURF6 diffused through the shell phase at low concentration and accumulated in the core phase, with the fluorescence intensity ~3-fold greater in the core vs. shell phase (Figures 3D and 3E; Video S3). Likewise, to monitor $rRNA_{EC}$ accumulation, we added a 3-fold excess of $rRNA_{EC}$ to condensates initially formed with a low concentration of rRNA (Figure 3F). Similarly, the supplemental $rRNA_{EC}$ diffused through the shell phase and accumulated in the core phase, causing the core volume fraction to increase by ~2-fold within 1 h (Figures 3G, 3H, and S2F). Additionally, to confirm the importance of rRNA-dependent interactions in the formation of multiphase condensates, we treated condensates with RNase A and observed conversion of two dense phases (multiphase condensates) to a single dense phase, demonstrating rRNA-dependent multiphase condensate formation (Figures S2G and S2H). Collectively, these results indicate that the components of cores create a phase with greater surface tension than the shells, causing cores to reside within the shell phases. Further, our results show that the multiphase condensates are under thermodynamic control by dynamically responding to changes in component concentrations.

We next investigated the origins of the apparent core-shell surface tension differences, which we hypothesized are due to differences in the hydrophobicity of the two phases, as we

(G) Pearson's coefficients for $rRNA_{EC}$ vs. NPM1 and $rRNA_{EC}$ vs. SURF6 calculated from the entire multiphase condensate (inset). Values represent mean ± SD, $n = 370$ condensates.
(H) NPM1, $rRNA_{EC}$, and SURF6 fluorescence intensity ratios inside the core and shell. Values represent mean ± SD, $n \geq 248$ condensates.
(I) Illustration of a multiphase condensate indicating enrichment of SURF6-$rRNA_{EC}$ in the core and NPM1 in the shell.
(J) Time-lapse micrographs of fusion events of shells and cores; scale bar, 1 μm. Shells fuse more rapidly than cores.
See also Video S1.

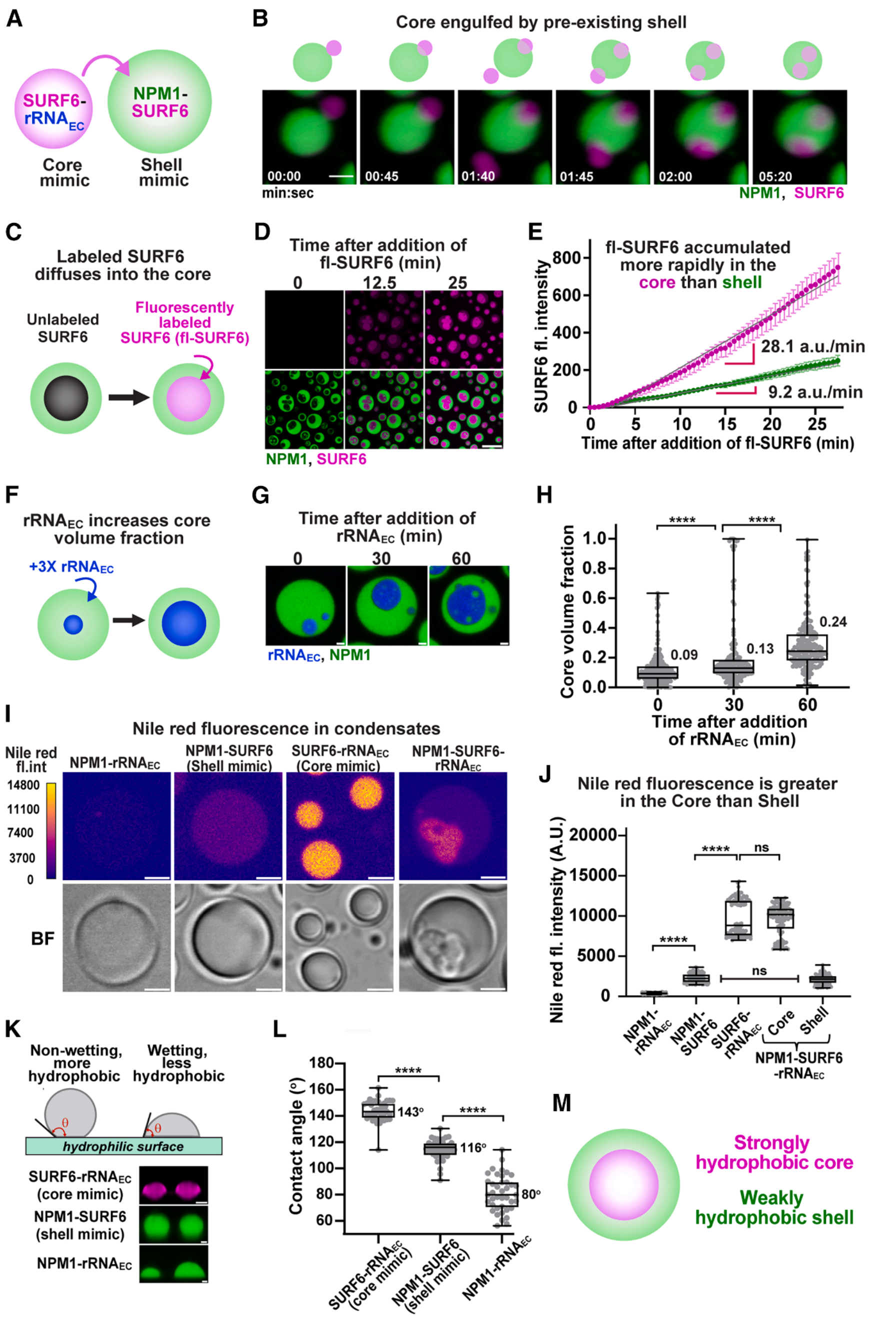

Figure 3. Emergence of core-shell architecture due to distinct physical properties of the nucleolar components

(A) Illustration of SURF6-rRNA$_{EC}$ condensates (core mimic) addition into preformed NPM1-SURF6 condensates (shell mimic).

(B) Time-lapse lattice light-sheet microscopy (LLSM) images showing engulfment of cores by pre-existing shells: NPM1 channel (green) and SURF6 channel (magenta); scale bar: 2 μm. See Video S2.

(C) Schematic of fluorescently labeled SURF6-AF647 (fl-SURF6) accumulation into the cores of preformed multiphase condensates containing NPM1-AF488 and unlabeled SURF6.

(D) Time-lapse micrographs of fl-SURF6 accumulation into the cores of multiphase condensates. Top row (SURF6 channel) and bottom row (overlay of the NPM1 and SURF6 channels); scale bar, 10 μm. See Video S3.

(E) Plot of fluorescence intensity of fl-SURF6 as a function of time, showing the rate of SURF6 accumulation in cores and shells. Values represent mean ± SD, $n = 12$ condensates.

(F) Schematic of increased core volume fraction upon excess rRNA$_{EC}$ addition to multiphase condensates.

(G) Fluorescence micrographs of NPM1 and rRNA$_{EC}$ in multiphase condensates forming larger cores as rRNA concentration increases; scale bar, 1 μm.

(H) Plot of the increase in volume fraction of cores over time after adding a 3-fold excess of rRNA$_{EC}$, $n \geq 160$ condensates; a two-tailed Mann-Whitney test was applied (****, $p < 0.0001$).

(I) Fluorescence micrograph of condensates stained with Nile red dye pseudocolored with the mpl-plasma LUT (top) and bright-field images (bottom) for binary (NPM1-rRNA$_{EC}$, NPM1-SURF6, and SURF6-rRNA$_{EC}$) and ternary (NPM1-SURF6-rRNA$_{EC}$) condensates; scale bar, 2 μm.

(J) Quantification of Nile red fluorescence intensity for binary and ternary condensates, $n = 100$; a two-tailed Mann-Whitney test was applied (****, $p < 0.0001$).

(K) Schematic for contact angle measurements (top). XZ projections of LLSM images for binary condensates on coverslips with a hydrophilic coating (bottom); scale bar, 2 μm.

(L) Contact angles measured for binary condensates on a hydrophilic surface, $n = 40$ condensates; a two-tailed Mann-Whitney test was applied; (****, $p < 0.0001$). Box plots in (H), (J), and (L) show the median (center line), 25th–75th percentiles (box), and minimum–maximum values (whiskers).

(M) Illustration of distinct physical properties of the core and shell phases of multiphase condensates.

showed previously for the DFC and GC[13] and preliminarily showed for NPM1/SURF6-N condensates.[20] To probe the physicochemical character of the core and shell phases, we used 8-anilinonaphthalene-1-sulfonic acid (ANS), a fluorescent dye that binds to hydrophobic pockets in partially folded molten globule proteins[34,35] and is also known to bind Arg and Lys residues in proteins through ion pair interactions.[36,37] Fluorescence microscopy and bright-field microscopy (Figure S2I, top and bottom, respectively) of various combinations of unlabeled NPM1, SURF6, and rRNA$_{EC}$ revealed significantly enhanced ANS fluorescence in the cores compared with shells of ternary multiphase condensates and in binary core vs. shell mimic condensates. ANS fluorescence intensity was greatest in multiphase condensate cores and binary core mimic condensates, which are enriched in SURF6 and rRNA$_{EC}$, and significantly lower in the corresponding shell phases, which are enriched in NPM1 (Figures S2I and S2J). Notably, the core exhibits ~2-fold higher hydrophobicity compared with the shell (Figure S2K). Due to the

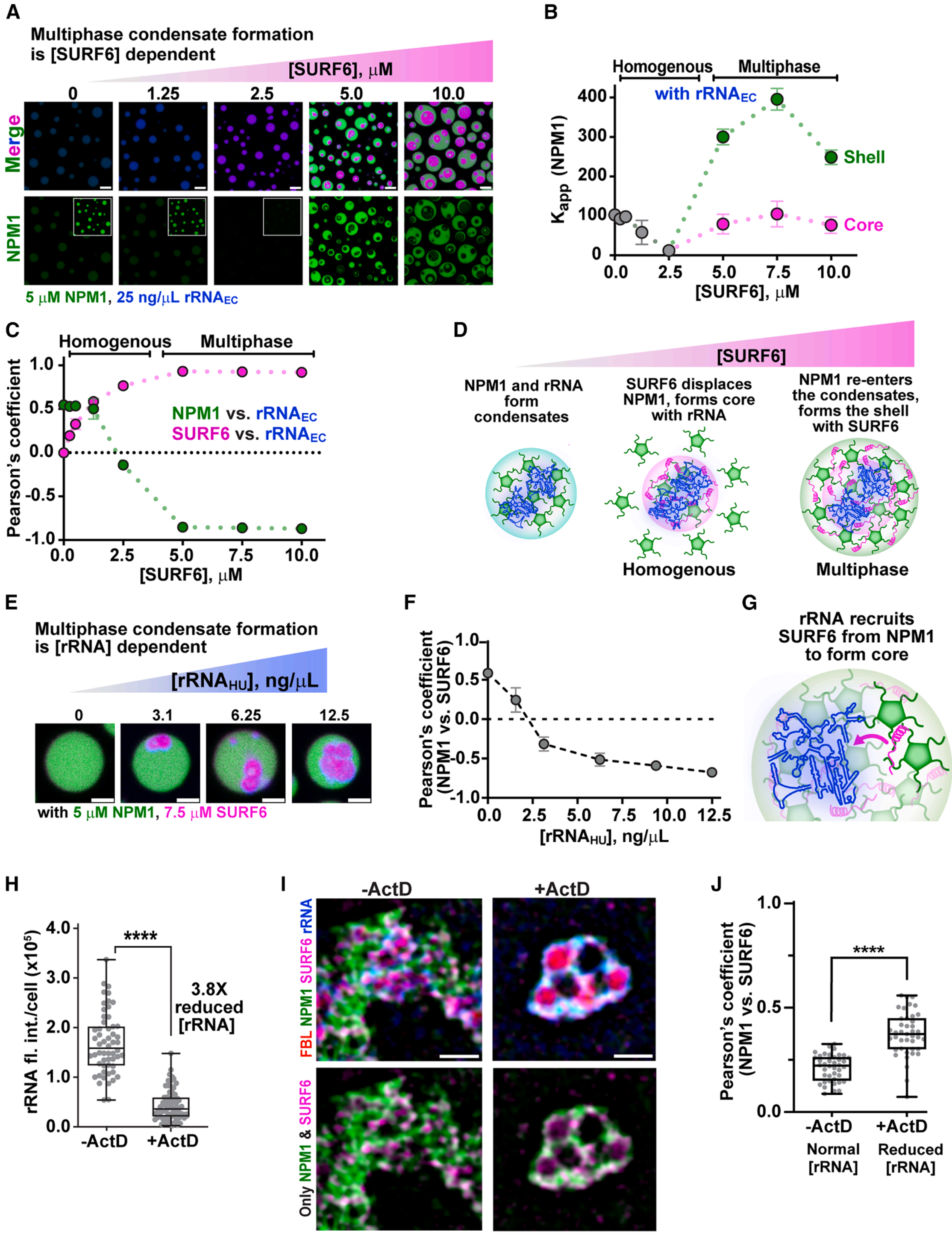

Figure 4. Concentration-dependent effect of SURF6 and rRNA on multiphase condensate formation and spatial heterogeneity within the GC

(A) Fluorescence micrograph of condensates formed upon titration of SURF6 into preformed NPM1 and $rRNA_{EC}$ condensates; scale bar, 5 µm. Fluorescence intensity threshold for the inset images in the bottom row was reduced 4-fold. See STAR Methods.

(B) Partition coefficient (K_{app}) for NPM1 as a function of increasing SURF6 concentration.

(C) Pearson's coefficients for NPM1/SURF6 vs. $rRNA_{EC}$ shown as a function of SURF6 concentration. Values for panels B and C represent mean ± SD, $n \geq 107$ condensates.

(D) Schematic of multiphase condensate formation mechanism with increasing SURF6 concentration.

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possibility of confounding effects of ANS binding to Arg and Lys residues through electrostatic interactions, we used Nile red as an alternative hydrophobicity probe. Nile red specifically binds to hydrophobic protein regions and does not interact with charged amino acids such as Arg or Lys, thereby providing an unbiased assessment of hydrophobic environments within the condensates. Nile red fluorescence intensity was greatest in multiphase condensate cores and binary core mimic condensates, and significantly lower in the corresponding shell phases (Figures 3I and 3J), consistent with the ANS results. We next investigated the surface properties of these condensates using lattice light-sheet microscopy to measure their contact angles on a hydrophilic surface (Figure 3K), with the expectation that hydrophobic condensates would resist wetting. Consistent with the results using ANS and Nile red dyes, the mean contact angle was greatest for core mimic condensates (143º), with shell mimic condensates exhibiting smaller contact angles (116º) and NPM1-$rRNA_{EC}$ condensates displaying the smallest contact angles (80º) (Figure 3L). Together, the ANS dye, Nile red dye, and surface wetting results indicate that the core phases possess a hydrophobic character (Figure 3M) and have relatively high surface tension and, thus, are energetically favored to reside within the interior of multiphase condensates.

Dynamic interplay between NPM1, SURF6, and rRNA *in vitro* and in cells

Our previous "hand-off" model proposed that NPM1 facilitates ribosome biogenesis by independently undergoing phase separation with rRNA and ribosomal proteins, which enter the GC via opposing fluxes, creating a liquid-like environment for ribosomal subunit assembly.[17,18] Here, we revisited this model in light of our findings, which show that the GC protein SURF6 preferentially binds $rRNA_{EC/HU}$ in the presence of NPM1. We first titrated SURF6 into mixtures with fixed concentrations of NPM1 (5 μM) and $rRNA_{EC/HU}$ (EC: 25 ng/μL and HU: 12.5 ng/μL) (Figures 4A and S3A). At SURF6 concentrations of up to 2.5 μM, homogeneous ternary condensates were observed. However, above this concentration, the condensates decomposed into two dense phases with core-shell architecture (Figures 4A, upper row, and S3A, upper row). Up to 2.5 μM, SURF6 progressively displaced NPM1 from the homogeneous condensates, decreasing the apparent NPM1 partition coefficient (K_{app}; Figures 4A, lower row, 4B, S3A, lower row, and S3B). Above 2.5 μM, most SURF6 formed the core phase with rRNA, while a small portion outside the cores underwent phase separation with NPM1 to form the shell phase. As the SURF6 concentration increased, NPM1 became increasingly delocalized from $rRNA_{EC/HU}$ (Figures 4C and S3C). These results show that SURF6 preferentially interacts with $rRNA_{EC/HU}$, forming a hydrophobic phase that is incompatible with the surrounding $rRNA_{EC/HU}$-depleted, hydrophilic NPM1-rich shell phase (Figure 4D).

To gain insight into the effect of rRNA concentration on multiphase condensate formation, we next titrated $rRNA_{HU/EC}$ into homogeneous condensates formed with NPM1 (5 μM) and SURF6 (HU: 7.5 μM and EC: 5 μM). Upon first addition, $rRNA_{HU/EC}$ was sequestered by SURF6 within core phases, which increased in size as the $rRNA_{HU/EC}$ concentration was increased (Figures 4E, S3D, and S3F). As seen with the titration of SURF6, NPM1 became increasingly delocalized from the SURF6-rich core phase within multiphase condensates as the $rRNA_{HU/EC}$ concentration increased (Figures 4F and S3E). These results demonstrate that SURF6 has a high capacity to sequester nascent rRNA within a dense, liquid-like hydrophobic phase in the presence of equimolar NPM1 (Figure 4G).

Our *in vitro* results showed that interactions between SURF6 and NPM1 within condensates are regulated by $rRNA_{HU/EC}$, with the two proteins co-localized at low $rRNA_{HU/EC}$ concentrations. To explore this regulatory interplay in cells, we treated DLD-1$^{NPM1\text{-}G/FBL\text{-}R}$ cells with Actinomycin D (ActD) to halt RNA polymerase I-dependent transcription[38–40] and reduce the level of rRNA (Figures 4H and S3G). Although treatment with ActD is well known to induce the formation of so-called nucleolar caps composed of FC and DFC components,[41] the effects of this treatment on the remaining GC components were unknown. In the absence of ActD treatment, NPM1 and SURF6 were partially spatially delocalized within the GC of DLD-1$^{NPM1\text{-}G/FBL\text{-}R}$ cells (Figures 4I [left], 4J, S3H [top], and S3I [top]). However, when rRNA was reduced through ActD treatment, spatial colocalization of NPM1 and SURF6 increased, as quantified using the PCC (Figures 4I [right], 4J, S3H [bottom], and S3I [bottom]). As was observed *in vitro*, the PCC value for NPM1 and SURF6 colocalization reached higher values at lower rRNA concentrations. Notably, although GC heterogeneity is reduced, it persists even upon ActD treatment, suggesting that nucleolar components in addition to NPM1, SURF6, and rRNA contribute to the GC's namesake granularity. The rRNA-dependent regulation of NPM1-SURF6 interactions represents one mechanism by which the GC's organization may be established and maintained. Together, our *in vitro* and cellular results demonstrate that the dynamic interplay between NPM1, SURF6, and rRNA contributes to the spatial heterogeneity within the GC. Next, we

(E) Fluorescence micrograph of condensates formed upon titration of additional $rRNA_{HU}$ into preformed condensates containing NPM1 and SURF6; scale bar, 2 μm.

(F) Pearson's coefficients for NPM1 vs. SURF6 as a function of increasing $rRNA_{HU}$ concentration. Values represent mean ± SD, $n \geq 122$ condensates.

(G) Schematic illustrating multiphase condensate formation with increasing rRNA concentration, leading to larger SURF6-rRNA cores as rRNA recruits more SURF6 into the cores.

(H) Sum of nucleolar rRNA fluorescence intensities quantified per cell in actinomycin D (ActD)-treated ($n = 70$) and untreated ($n = 56$) DLD-1$^{NPM1\text{-}G/FBL\text{-}R}$ cells; a two-tailed Mann-Whitney test was applied; (****, $p < 0.0001$).

(I) SIM images of untreated (left) and ActD-treated (right) DLD-1$^{NPM1\text{-}G/FBL\text{-}R}$ cells; scale bar, 1 μm. The top row of images shows overlaid channels of FBL, NPM1, SURF6, and rRNA, whereas the bottom row shows only the NPM1 and SURF6 channels.

(J) Pearson's coefficients for NPM1 vs. SURF6 for ActD-treated ($n = 41$) and untreated ($n = 40$) cells; a two-tailed Mann-Whitney test was applied; (****, $p < 0.0001$). Box plots in (H) and (J) show the median (center line), 25th–75th percentiles (box), and minimum–maximum values (whiskers)

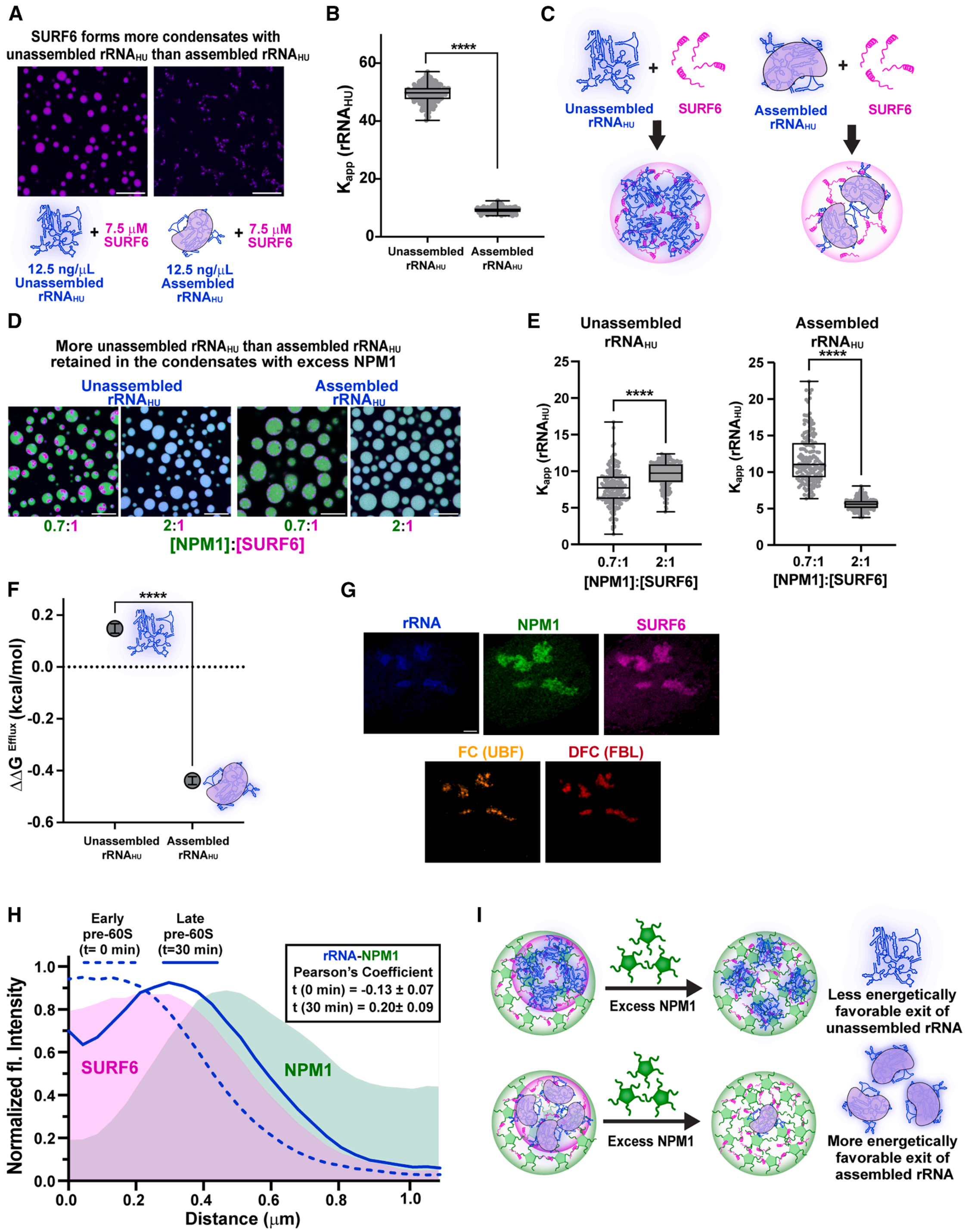

Figure 5. Dynamic interplay between nucleolar components in the GC regulates the outward flux of ribosomal subunits

(A) Fluorescence micrograph of condensates composed of SURF6 with unassembled $rRNA_{HU}$ (total human rRNA, left) or assembled $rRNA_{HU}$ (human pre-60S particles, right) in TBS-150 containing 3 mM $MgCl_2$. Unassembled $rRNA_{HU}$ and assembled $rRNA_{HU}$ are stained with SYTO 40; scale bar, 10 µm.

(B) Comparison of the partition coefficient (K_{app}) values for $rRNA_{HU}$ in the condensates formed in panel A, $n = 222$ and 169 condensates for unassembled $rRNA_{HU}$ and assembled $rRNA_{HU}$, respectively; a two-tailed Mann-Whitney test was applied; (****, $p < 0.0001$).

(C) Schematic of reduced partitioning of assembled vs. unassembled $rRNA_{HU}$ into SURF6-$rRNA_{HU}$ condensates.

(D) Fluorescence micrographs of condensates composed of NPM1, SURF6, and unassembled $rRNA_{HU}$ (left) or assembled $rRNA_{HU}$ (right) at 0.7:1 and 2:1 concentration ratios of NPM1:SURF6; scale bar, 10 µm.

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investigated the implications of this dynamic interplay on the efflux of assembled ribosomal subunits from the GC.

Dynamic interplay between nucleolar components in the GC regulates the outward flux of ribosomal subunits

We previously showed that efflux of $rRNA_{EC}$ within assembled ribosomal subunits from the GC is facilitated by reduced thermodynamic favorability of interactions with NPM1 that underlie phase separation.[18] As $rRNA_{EC}$ assembles with ribosomal proteins within subunits, interaction sites are buried, weakening interactions with NPM1. However, at the time of this report, the role of SURF6/rRNA interactions in creating the heterogeneous structure of the GC and their implications for the outward flux of assembled ribosomal subunits were not appreciated. Therefore, we revisited this "flux model" through the lens of multiphase NPM1/SURF6/rRNA condensates formed using human nucleolar components. We first demonstrated that SURF6 exhibits a greater propensity to phase separate with unassembled, nascent $rRNA_{HU}$ (total human rRNA) compared with assembled $rRNA_{HU}$ (human pre-60S ribosomal particles), forming more numerous, larger, and denser condensates with the unassembled species (Figures 5A–5C, S4A, and S4B). We then separately formed multiphase condensates containing NPM1, SURF6, and the two $rRNA_{HU}$ species, and subsequently added a stoichiometric excess of NPM1 to mimic the NPM1-abundant GC environment.[17,18] Although most unassembled $rRNA_{HU}$ remained in condensates despite elimination of the core-shell architecture (Figures 5D [left], 5E [left], and S4C), only 45% of assembled $rRNA_{HU}$ was retained (Figures 5D [right], 5E [right], and S4D). Based on K_{app} values, the efflux of assembled $rRNA_{HU}$ is significantly more thermodynamically favorable ($\Delta\Delta G^{Efflux}$ = –0.44 kcal/mol) than it is for unassembled $rRNA_{HU}$ ($\Delta\Delta G^{Efflux}$ = 0.15 kcal/mol; Figure 5F).

Remarkably, the bacterial system displayed analogous behavior. Similar phase separation behaviors were observed with unassembled $rRNA_{EC}$ and assembled $rRNA_{EC}$ (70S ribosomes) (Figures S5A and S5B), indicating conserved biophysical properties and interaction mechanisms between systems. Consistent with human results, 77% of unassembled $rRNA_{EC}$ was retained after excess NPM1 addition (Figures S5C, S5E [left], and S5F [left]), whereas only 28% of assembled $rRNA_{EC}$ remained (Figures S5D, S5E [right], and S5F [right]). The NPM1-mediated extraction of assembled $rRNA_{EC}$ was more thermodynamically favorable ($\Delta\Delta G^{Efflux}$ = –0.43 kcal/mol) than for unassembled $rRNA_{EC}$ ($\Delta\Delta G^{Efflux}$ = –0.12 kcal/mol; Figure S5G). Furthermore, the lower SURF6 correlation with assembled $rRNA_{EC}$ in homogeneous vs. multiphase condensates (Figure S5H) demonstrates that NPM1 effectively extracted assembled $rRNA_{EC}$ from the SURF6-rich core. Additionally, the mean SURF6 intensity was reduced in condensates containing assembled $rRNA_{EC}$ (Figure S5I), indicating SURF6 retention rather than complete exclusion from the condensates.

Alternatively, to test more favorable core enrichment of unassembled $rRNA_{EC}$, we formed homogeneous ternary condensates of NPM1, SURF6, and assembled $rRNA_{EC}$ (Figure S5J), then titrated unassembled $rRNA_{EC}$ (6.25, 12.5, and 25 ng/μL), followed by an additional 5 μM SURF6 to induce multiphase condensates. We observed a decrease in assembled $rRNA_{EC}$ intensity within cores as a function of increasing unassembled $rRNA_{EC}$ concentration (Figure S5K), supporting the idea that differential partitioning of rRNA species drives spatial organization in the nucleolar GC. To further validate these findings in cells, we included an additional, extended 30-min chase time point representing more mature rRNA species in our nucleolar imaging studies (Figure 5G) and performed RDF analysis (Figure 5H). Interestingly, we observed an increase in the rRNA-NPM1 correlation (over the 0-min chase time point, Figure 1G), supporting our *in vitro* results (Figures 5H, inset, S4E, and S4F). Together, these findings highlight how the interplay between SURF6 and NPM1 mediates the retention of nascent, unassembled $rRNA_{HU/EC}$ within GC-like multiphase condensates while gating efflux of mature, assembled $rRNA_{HU/EC}$ (Figure 5I).

DISCUSSION

Our data show that structural heterogeneity in the GC arises from liquid-like sub-phases marked by the differential localization of NPM1, SURF6, and rRNA (Figures 6A–6C). SURF6, a positively charged early pre-60S assembly factor, preferentially phase separates with nascent rRNA, confining it within an interior GC sub-phase. SURF6, composed of 13% Lys and 15% Arg residues, initially displaces NPM1 from homogeneous NPM1/rRNA condensates, forming a dense SURF6/rRNA-rich phase. However, due to NPM1's capacity for multivalent interactions, it subsequently phase separates with residual SURF6, forming an outer NPM1-enriched shell that encapsulates SURF6/rRNA cores. The core-shell architecture reflects fundamental physicochemical differences between SURF6 and NPM1, with the high Arg content of SURF6 creating hydrophobic cores that are immiscible with the hydrophilic, NPM1-rich shell phase. Interestingly, the

(E) Plot of partition coefficient (K_{app}) values for unassembled $rRNA_{EC}$ (left) and assembled $rRNA_{EC}$ (right) at 0.7:1 and 2:1 concentration ratios of NPM1:SURF6; $n \geq$ 164 condensates; a two-tailed Mann-Whitney test was applied (****, $p < 0.0001$). Box plots in (B) and (E) show the median (center line), 25th–75th percentiles (box), and minimum–maximum values (whiskers).

(F) Calculated Gibbs free energies of particle efflux ($\Delta\Delta G^{Efflux}$) of unassembled $rRNA_{HU}$ and assembled $rRNA_{HU}$ from the multiphase condensates from panel E upon the addition of excess NPM1. Values represent mean ± SEM. Welch's unpaired *t* test was applied (****, $p < 0.0001$).

(G) Super-resolution Airyscan image of a DLD-1$^{NPM1\text{-}G/FBL\text{-}R}$ cell nucleolus immunostained for SURF6, UBF, NPM1-mEGFP, mCherry-FBL, and EU-rRNA chased after 30 min (pulse-EU labeled for 30 min followed by a 30-min chase) labeled with AF647; scale bar, 2 μm. See STAR Methods.

(H) Min-max-normalized intensities for SURF6 (magenta trace), NPM1 (green trace), and EU-rRNA (pulse-EU labeled for 30 min without chase [dashed blue trace] and after 30 min chase [solid blue trace]) by distance from the FC center. Pearson's coefficients for rRNA vs. NPM1 (0-min chase and 30-min chase) are indicated in the inset. Values represent mean ± SEM, n = 41 cells and 33 cells for t = 0 min and t = 30 min, respectively.

(I) Schematic depicting the more favorable efflux of assembled rRNA from multiphase condensates upon addition of excess NPM1 due to weakened interactions with SURF6.

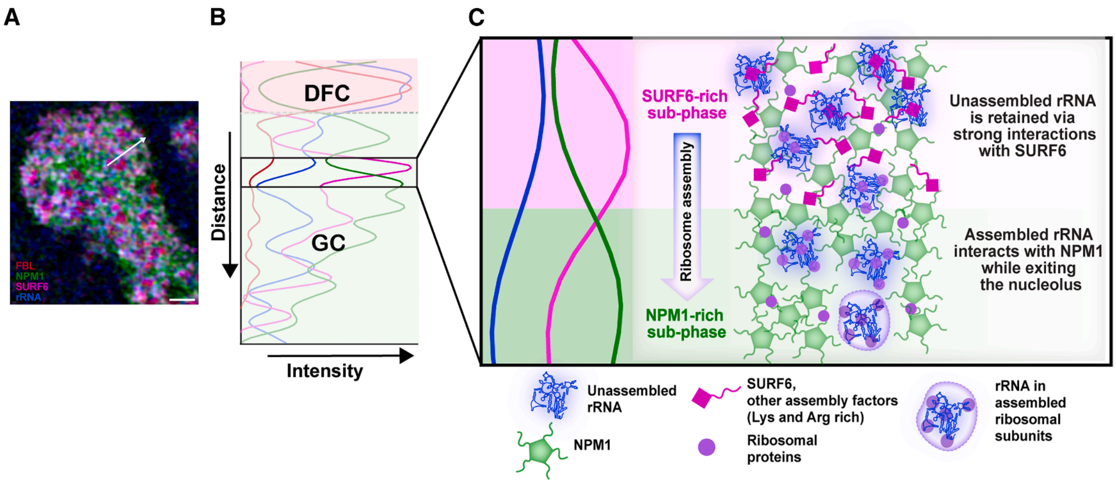

Figure 6. Structural heterogeneity in the GC of the nucleolus underlies ribosomal subunit assembly

Schematic of SURF6-rich and NPM1-rich liquid-like sub-phases within the nucleolar GC.

(A) SIM image of a nucleolus.

(B) Normalized fluorescence intensity plot along the white arrow in (A), with the DFC and GC regions indicated.

(C) The boxed region of the GC in (B) is expanded (left) to illustrate the spatial delocalization of SURF6 and rRNA vs. NPM1. Proposed model (right): as ribosomal subunits assemble, they successively move outward (downward in the illustration) through SURF6-rich and NPM1-rich sub-phases under thermodynamic control, driven by differential interactions of these proteins with rRNA across different states of ribosomal subunit assembly. SURF6 prefers to bind rRNA in early assembly states, which enables highly abundant NPM1 to extract assembled ribosomal subunits from interior SURF6-rich sub-phases.

Arg-rich peptide protamine also forms viscoelastic, hydrophobic condensates under crowded conditions.[42] In contrast, basic tracts in NPM1's IDR are Lys-rich (24 Lys and 3 Arg residues),[17,20] accounting for the hydrophilic character of the NPM1-rich shell phase. The Arg-richness of SURF6 also enables it to out-compete Lys-rich NPM1 for rRNA binding, as has also been observed for Arg and Lys polymers.[43] These observations suggest that the Arg/Lys balance in SURF6 and NPM1 has evolved to establish distinct GC sub-phases. High affinity for rRNA and its hydrophobic character drive the preferential localization of SURF6 with nascent rRNA in the interior, providing a mechanistic basis for the sub-compartmentalization of the nucleolar GC.

SURF6, while undergoing phase separation in the GC, also acts as an assembly factor during pre-60S ribosome biogenesis.[15,44] In this latter role, the C terminus of SURF6 (residues 212–361) adopts an α-helical structure upon binding, together with SSF1 and RRP15, to a specific region of rRNA within states A and B during the assembly of pre-60S particles.[15] These interactions organize 28S rRNA helices before being displaced as assembly progresses.[15] Based on our mechanistic data, we propose that SURF6's Arg- and Lys-richness enables retention within the GC through phase separation as it cycles on and off assembling pre-60S particles. Importantly, the intrinsically disordered N-terminal region of SURF6 (residues 1–211), unresolved in structures of early assembly states,[15] is always available for multivalent interactions with immature rRNA species within the GC. Interestingly, two C-terminal regions of SSF1 (residues 205–265 and 297–473) are also structurally unresolved and likely disordered in the early assembly states, with the latter region enriched in Arg and Lys residues, similar to SURF6 (Figures S6A and S6B). Also, the N- and C-terminal regions of RRP15 are structurally unresolved (residues 1–126 and 212–282), and both are enriched in Asp and Glu residues interspersed with Arg and Lys residues (Figures S6A and S6B). Our cellular and *in vitro* data support the conclusion that SSF1 (Figures S6C–S6I), similar to SURF6, cycles on and off assembling pre-60S particles and is retained in the GC interior through phase separation. Furthermore, bioinformatics analyses suggest that other structurally characterized pre-60S assembly factors behave similarly due to enrichment of Arg and Lys residues within IDRs in their structures (Figures S6J and S6K). We propose that these positively charged IDRs enable many pre-60S assembly factors to be retained in the interior of the liquid-like GC through multivalent interactions with rRNA that promote phase separation. As assembly progresses, these factors adopt folded structures within specific pre-60S states before being released but spatially retained. Our observations highlight the synergy between constitutively disordered and conditionally folded regions of assembly factors that contribute to the liquid-like nature of the GC and facilitate the sequential assembly of specific structures within 60S ribosomal subunits.

We show that NPM1 and the assembly factors SURF6 and SSF1 differentially localize with rRNA in the GC, reflecting this nucleolar sub-compartment's namesake granularity (Figures S6G–S6I). We propose that other assembly factors with properties similar to those of SURF6, discussed above, also contribute to this granularity. The principles we describe underlying differential interactions of SURF6 and NPM1 with rRNA at different stages of pre-60S assembly likely extend to other assembly factors, collectively contributing to GC heterogeneity. The affinity of SURF6 for rRNA within the GC interior declines as ribosomal subunits assemble, enabling hand-offs to highly abundant, peripheral NPM1.[18,21,45] SURF6 is 50-fold less abundant in human cells than NPM1[45] but is enriched in the GC interior through high-affinity interactions and phase separation with nascent rRNA species. We previously showed that NPM1's affinity for ribosomal subunits declines as they mature and diffuse outward from the GC,[18] ultimately escaping into the nucleoplasm. Results in yeast showed that some NPM1 (Fkbp39 in *S. pombe*) remains bound nonspecifically to nucleoplasmic pre-60S particles[46] while most is retained in the GC due to the constant flux of assembling ribosomal subunits in healthy cells. Thus, the dynamic interplay between SURF6, additional SURF6-like assembly factors, and NPM1 mediates the assembly of ribosomal subunits within and efflux from the nucleolar GC. Additionally, we speculate that the condensation-based interplay between SURF6 and NPM1 may adaptively engage aberrant rRNA species to manage ribosome assembly defects and maintain cellular homeostasis. However, further study is needed to test this hypothesis. In conclusion, our findings broadly illustrate how the formation of phase-separated mesoscale structures orchestrates critical steps, e.g., ribosome subunit assembly, in the essential biological process of ribosome biogenesis.

Limitations of the study

We employed super-resolution microscopy to study the nucleolar GC, which revealed spatial heterogeneity of rRNA and SURF6 relative to NPM1. However, the spatial resolution inherent to this microscopy method is insufficient for ultrastructural interpretation at the nanometer scale. In the future, cryo-electron tomography could be used to study GC heterogeneity with greater spatial resolution. Further studies are also needed to determine whether additional assembly factors contribute to the multiphase nature of the GC as they participate in the subunit assembly process. Furthermore, in this study, we examined the GC's multiphase behavior only in DLD-1 cells, which were engineered to express fluorescently tagged endogenous NPM1 and FBL. Whether the GC in other cell types exhibits analogous multiphase behavior remains to be determined in future studies. Nonetheless, the fundamental biomolecular determinants of GC multiphase behavior recapitulate key aspects of our observations in DLD-1 cells. This supports our proposal that these biomolecules play similar roles across cell types.

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Dr. Richard W. Kriwacki (richard.kriwacki@stjude.org).

Materials availability

The cell lines and bacterial strains will be made available upon request. The plasmids generated in this study are available from Addgene (IDs: 255296, 255297, and 255298).

Data and code availability

- Source data and western blots are available in Data S1 and S2.
- The code for RDF analyses is available at Zenodo: https://doi.org/10.5281/zenodo.20331552. The code for extracting the unresolved protein regions from the cryo-electron microscopy (cryo-EM) structures of the human pre-60S assembly factors is available upon request.
- Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

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AUTHOR CONTRIBUTIONS

Conceptualization, P.D., M.C.F., and R.W.K.; methodology, P.D., M.C.F., Q.M., M.T., S.M.P.-M., S.T., D.W.B., J.J., K.J., and R.W.K.; investigation, P.D., M.C.F., S.K., Q.M., M.T., A.P., G.E.C., K.J., T.F., E.G., and C.-G.P.; visualization, P.D., M.C.F., S.K., Q.M., A.P., S.T., D.W.B., J.J., K.J., and R.W.K.; formal analysis, P.D., M.C.F., S.T., D.W.B., and J.J.; validation, P.D., M.C.F., S.K., Q.M., S.T., D.W.B., J.J., and K.J.; data curation, P.D., M.C.F., S.T., D.W.B., and J.J.; software, S.T., D.W.B., and J.J.; funding acquisition, R.W.K.; supervision, R.W.K.; writing – original draft, P.D. and R.W.K.; and writing – review and editing, P.D., M.C.F., S.K., Q.M., S.M.P.-M., A.P., S.T., D.W.B., J.J., G.E.C., K.J., R.B., and R.W.K.

DECLARATION OF INTERESTS

The authors declare no competing interests.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Anti-UBF (F9) mouse monoclonal conjugated to Alexa Fluor 647	Santa Cruz Biotechnology	Cat#: SC-13125AF647; RRID:AB_671403
Anti-SURF6 rabbit polyclonal	Invitrogen	Cat#: PA5-54841; RRID:AB_2648093
Anti-GFP nanobody conjugated to Alexa Fluor 488	Chromotek	Cat#: GB2AF488; RRID:AB_2827573
Anti-RFP nanobody conjugated to Alexa Fluor 568	Chromotek	Cat#: RB2AF568; RRID:AB_2827576
Anti-PPAN rabbit polyclonal	Proteintech	Cat#: 11006-1-AP; RRID: AB_2168040
Anti-DDX18 rabbit polyclonal	GeneTex	Cat#: GTX103392; RRID: AB_2885240
DyLight 405 AffiniPure F(ab')$_2$ Fragment Donkey Anti-Mouse IgG (H+L)	Jackson ImmunoResearch	Cat#: 715-476-151; RRID: AB_2632566
Bacterial and virus strains		
BL21 (DE3) E. coli	MilliporeSigma	Cat#: 69450
K12 A19 E. coli	Kriwacki Lab, St. Jude	N/A
Chemicals, peptides, and recombinant proteins		
Alexa Fluor 647 Azide, Triethylammonium Salt	Thermo Fisher	Cat#: A10277
The Click-It RNA Imaging Kit	Invitrogen	Cat#: C10330
Actinomycin D	Thermo Fisher	Cat#: A7592
Poly-L lysine solution	Sigma Aldrich	Cat#: P8920-100mL
RPM1 1640 medium	Gibco	Cat#: A10491-01
Fetal Bovine Serum	Gibco	Cat#: 26140079
Blocker BSA (10%) in PBS	Thermo Scientific	Cat#: 37525
Paraformaldehyde	Electron Microscopy Sciences	Cat#: 15710
Dulbecco's Phosphate Buffered Saline	Gibco	Cat#:14190-144
ProLong Diamond Antifade Mountant	Thermo Fisher	Cat#: P36961
STYO™ 40 blue fluorescent nucleic acid stain	Life Technologies, Thermo Fisher Scientific	Cat#: S11351
Alexa Fluor 488 C5 Maleimide	Invitrogen by Thermo Fisher Scientific	Cat#: A10254
Alexa Fluor 647 C2 Maleimide	Invitrogen by Thermo Fisher Scientific	Cat#: A20347
Alexa Fluor 555 NHS Ester	Invitrogen by Thermo Fisher Scientific	Cat#: A20009
Nile red	Sigma Aldrich	Cat#: 72485
RNase A	Sigma Aldrich	Cat#: 70856
Protease Inhibitor	Sigma Aldrich	Cat#: S8830
8-Anilino-1-naphthalenesulfonic acid ammonium salt (ANS)	Sigma Aldrich	Cat#: 28836-03-5
Penicillin-streptomycin	ThermoFisher Scientific	Cat. #15140122
e-Myco PLUS Mycoplasma PCR Detection Kit	Bulldog Bio	Cat. #2523448
sgRNA (CAGE117.NPM1.g1)	Synthego	5′-UCCAGGCUAUUCAAGAUCUC-3′
NLS SpCas9 protein	St. Jude Protein Production Core	N/A
PowerPlex® Fusion System	Promega	Cat#:DC2402
His-NPM1 (wild-type)	Mitrea et al.[21]	N/A
His-NPM1 (C21T/C275T)	White et al.[47]	N/A
His-SURF6 (wild-type)	This paper	Addgene #255296

(Continued on next page)

Continued		
REAGENT or RESOURCE	SOURCE	IDENTIFIER
His-SURF6 (C19S)	This paper	Addgene #255297
His-SSF1	This paper	Addgene #255298
TEV protease	Kriwacki Lab, St. Jude	N/A
Deposited data		
Code for radial distribution function (RDF) analyses	This paper	Zenodo: https://doi.org/10.5281/zenodo.20331552
Experimental models: Cell lines		
Human: DLD-1 cells	ATCC	Cat#: CCL-221; RRID:CVCL_0248
Human: DLD-1$^{NPM1\text{-}G/FBL\text{-}R}$ cells	This paper	N/A
Software and algorithms		
Imaris	Oxford Instruments	https://imaris.oxinst.com/
FIJI	NIH	https://fiji.sc/
Slidebook 6	Intelligent Imaging Innovations, Inc., Denver, CO	https://www.intelligent-imaging.com/slidebook
Arivis Vision Pro	Zeiss, Jena, Germany	https://www.zeiss.com/microscopy/us/products/software/arivis-pro.html
Biopython	http://biopython.org	RRID:SCR_007173
cif-to-fasta	https://github.com/shin-kinos/cif-to-fasta	N/A
GraphPad Prism 10	GraphPad Software	https://www.graphpad.com/scientific-software/prism/

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Cell lines

DLD-1 (male, adult, age not reported, Dukes' type C colon cancer) cell lines were purchased from American Type Culture Collection (ATCC). DLD-1 cells were maintained in RPMI 1640 medium (Thermo Fisher Scientific), supplemented with 10% fetal bovine serum and 100 U/mL penicillin-streptomycin. Cultures were incubated at 37 °C in a humidified 5% CO_2 environment. Gene-edited lines underwent authentication through short tandem repeat (STR) profiling, and mycoplasma contamination was ruled out using the e-Myco PLUS Mycoplasma PCR Detection Kit (bulldog Bio).

Generation of endogenously NPM1 and FBL-tagged cell line

DLD1-NPM1-mEGFP/FBL-mCherry C-terminally double-tagged (DLD-1$^{NPM1\text{-}G/FBL\text{-}R}$) cells were generated using CRISPR-Cas9 technology in the Center for Advanced Genome Engineering (St. Jude), as described.[31] Briefly, the DLD-1$^{NPM1\text{-}G/FBL\text{-}R}$ cell line was created by transiently co-transfected 500,000 DLD1 cells with pre-complexed ribonuclear proteins (RNPs) consisting of 100 pmol of chemically modified sgRNA (CAGE117.NPM1.g1, Synthego), 33 pmol of 3X NLS *Sp*Cas9 protein (St. Jude Protein Production Core), and 500 ng of plasmid donor (Biobasic USA). The transfection was performed via nucleofection (Lonza, 4D-Nucleofector™ X-unit) using solution P3 and program CA-137 in a small (20 µL) cuvette according to the manufacturer's recommended protocol. Single cells were sorted for GFP positivity five days post-nucleofection into 96-well plates containing prewarmed media and clonally expanded. Clones were screened and verified for the desired modification using PCR-based assays (5′ junction primers – CAGE117.gen.F2 and CAGE117.junc.meGFP.DS.R2, 3′ junction primers – CAGE117.junc.meGFP.DS.F2 and CAGE117.gen.R2, zygosity confirmation primers – CAGE117.DS.F and CAGE117.DS.R) and confirmed via sequencing. A resulting *NPM1_mEGFP* tagged clone (2C4) was then transiently co-transfected with CAGE223.FBL.g3, 3X NLS *Sp*Cas9 protein, and CAGE223.g3.mCherry.donor as described above. Additionally, 1 µM of Nedisertib (M3814) was added to the media for 24 hours after nucleofection. Cells were single-cell sorted for mEGFP and mCherry double-positive cells and clonally expanded. Clones were screened and verified for the desired targeted integration using PCR-based assays (5′ junction primers – CAGE223.gen.F and CAGE223.junc.DS.R, 3′ junction primers – CAGE223.junc.DS.F and CAGE223.gen.R, zygosity confirmation primers – CAGE223.FBL.DS.F and CAGE223.FBL.DS.R) followed by sequence confirmation. All final clones tested negative for mycoplasma by the MycoAlert™Plus Mycoplasma Detection Kit (Lonza) and were authenticated using the PowerPlex® Fusion System (Promega) performed at the Hartwell Center (St. Jude). Editing construct sequences and screening primers are listed in Table S1.

Bacterial strain

E. coli BL21 (DE3) cells were purchased from MilliporeSigma and used for protein expression. The cells were grown in Lauria Broth. Protein expression was induced at an OD_{600} ~0.8 with 0.4 mM IPTG. Cultures were incubated using an InnOva 4230 incubator/shaker. The cells were harvested by centrifugation and pellets stored at −30 °C.

METHOD DETAILS

EU labeling and immunostaining

DLD-1$^{NPM1\text{-}G/FBL\text{-}R}$ cells were grown to approximately 50% confluence on a circular 12 mm diameter #1.5 coverslips (Electron Microscopy Sciences, cat#:72230-01) and treated with 1 mM 5-ethynyl uridine (EU) for 30 minutes. For t=0, EU-RNA label chase time, cells were immediately fixed for 10 minutes using 4% paraformaldehyde (PFA) in PBS. To follow EU-labeled rRNA (t = 30 min), media was replaced with media without EU and incubated for another 30 minutes and then fixed with PFA. Following fixation, coverslips were washed twice with Dulbecco's phosphate-buffered saline (DPBS) buffer. Cells were permeabilized with 0.5% Triton X-100 in PBS for 5 mins. EU incorporated into newly synthesized RNA was fluorescently labeled with Alexa-647 fluorophore using an RNA Click-It RNA imaging kit (Molecular Probes) following the manufacturer's protocol (https://assets.thermofisher.com/TFS-Assets/LSG/manuals/mp10329 .pdf). For immunostaining, coverslips were initially blocked with 4% bovine serum albumin (BSA) in DPBS. To probe for SURF6 or SSF1, we used an anti-SURF6 rabbit polyclonal antibody (Invitrogen, PA5-54841) and anti-PPAN (SSF1) rabbit polyclonal antibody (11006-1-AP, Proteintech) at 1:250 dilution. To obtain good-quality SIM imaging, we enhanced the signals of mEGFP in NPM1 and mCherry in fibrillarin using fluorophore-conjugated nanobodies. We accomplished this by adding GFP nanobodies conjugated to Alexa 488 (GB2AF488, Chromotek) and RFP nanobodies conjugated to Alexa 568 (RB2AF568, Chromotek), to the SURF6 or SSF1 antibody mixture (1:500 dilution). Samples were incubated in the antibody mixture for an hour at room temperature.

For Airyscan imaging samples used for RDF analyses we also probed for UBF by adding anti-UBF mouse monoclonal antibodies (sc-13125, Santa Cruz Biotechnology) diluted 250-fold. We similarly used the GFP-Alexa 488 nanobodies but used RFP nanobodies conjugated to Atto 594 fluorophore (RBA594, Chromotek) for FBL to minimize fluorescence crosstalk. Bound SURF6 or SSF1 antibodies were probed using fluorophore-conjugated secondary antibody AffiniPure F(ab')$_2$ Fragment Donkey Anti-Rabbit IgG1-Dylight 405 (Jackson ImmunoResearch, 711476152) diluted to 1:400. We used Alexa 555 conjugated goat anti-mouse F(ab')$_2$ antibodies (4409S, Cell Signaling) diluted 1:400 as secondary antibodies for UBF. Coverslips were washed with DPBS and water and mounted on a glass slide (Electron Microscopy Sciences, 71863-01) using ProLong Glass Anti-fade Mountant (Invitrogen). Samples were allowed to set for at least 72 hours prior to imaging.

To probe for the three layers of the nucleolus, we immunostained fixed DLD-1$^{NPM1\text{-}G/FBL\text{-}R}$ cells with anti-GFP Alexa 488 (for NPM1) and anti-RFP Alexa 568 (for FBL) nanobodies (1:500 dilution) and anti-UBF conjugated to Alexa 647 (Santa Cruz Biotech, SC-13125) at 1:250 dilution. Mounted slides were similarly allowed to be set for 72 hours prior to imaging.

Actinomycin D treatment

DLD-1$^{NPM1\text{-}G/FBL\text{-}R}$ cells were seeded onto poly-Lysine coated coverslips to approximately 50% confluence and synchronized by overnight serum starvation. Media was replaced the following day, and cells were allowed to recover for two hours prior to 30 min treatment with 1 mM EU. Following this, cells were washed three times with pre-warmed media to remove residual EU. Post-wash, cells were incubated with media containing either 0.1 μg/mL actinomycin D (ActD) (Invitrogen) or DMSO for one hour. To minimize fixation artifacts, cells were directly fixed at the selected time point in media using 4% pre-warmed PFA.

SIM imaging and processing

Multicolor super-resolution SIM (SR-SIM) images were collected using a Zeiss ELYRA PS.1 super-resolution microscope (Carl Zeiss MicroImaging) using a 100X Plan-Apochromat oil objective lens with 1.46 NA. Images were acquired using 405 nm (Dylight 405, SURF6), 488 nm (GFP/Alexa 488, NPM1), 561 nm (RFP/Alexa 568, FBL), and 660 nm (Alexa 647, rRNA) laser lines. Five orientation angles of the excitation grid were acquired for each Z plane, where Z spacing is 90 nm between planes. SIM images were processed using the SIM analysis module of the Zen 2012 BLACK software (Carl Zeiss MicroImaging). To image the three layers of the nucleolus, we used the 488 nm (GFP/Alexa 488, NPM1), 561 nm (RFP/Alexa 568, FBL), and 660 nm (Alexa 647, UBF) laser lines.

For proper spatial alignment of multichannel SR-SIM images, a post-acquisition channel alignment was performed for all SR-SIM images. To generate the calibration file, 100 nm TetraSpeck beads were seeded onto a #1.5 coverslip, mounted to a slide, and imaged using the same imaging parameters as the sample. Data was reconstructed with the SIM analysis module before using the Channel Alignment processing function, selecting the "Affine" and "Fit" options within the Zen 2012 BLACK software. After the calibration file was generated, it was saved and utilized for subsequent imaging experiments. SR-SIM images were aligned using the calibration file generated in the Zen 2012 BLACK software using the Channel Alignment processing function.

Super-resolution Airyscan imaging and processing

Multicolor super-resolution Airyscan images were acquired using a Zeiss LSM 980 Airyscan 2 microscope using a 63X Plan-Apochromat oil objective with 1.4 NA. Raw images were processed using Zen BLUE software using Automatic Standard 3D processing option.

Colocalization analyses

SIM images of nucleoli were visualized and analyzed using IMARIS software (Oxford Instruments). Due to the highly heterogeneous spatial distribution of components and irregular shapes of nucleoli, we selected the best-focused single slice for analyses. For the colocalization of endogenous NPM1, rRNA, SURF6, and FBL, we subdivided the nucleolus into several ROIs that encompassed the DFC and the inner GC regions. We chose these areas based on the established role of SURF6 as an early assembly factor for 60S pre-ribosomal particles.[15] We reasoned that, since SURF6 binds to rRNA during the early maturation steps, focusing on regions surrounding the DFC and inner GC would give the highest probability to observe early assembly steps where NPM1 and SURF6 compete for rRNA binding. To establish the ROIs, we segmented the DFC regions of the nucleolus based on the intensities from the FBL channel and generated binary masks. We expanded the DFC masks using the dilation function of FIJI software. To segment the intensities for protein components (NPM1, SURF6, and FBL), we set a threshold to include only intensities above 25% of the maximum intensities. Due to the strong background and heterogeneous signals across samples for rRNA, we used thresholds at 15-40% of the maximum intensities for segmentation. We calculated the Mander's overlap coefficient for ROIs identified in six individual nucleoli, each with multiple DFC regions.

To examine the effect of ActD treatment on the spatial distribution of the nucleolar components, we analyzed their colocalization within the entire nucleolus. To define the nucleolus, we segmented ROIs using the total intensity of NPM1, SURF6, rRNA, and FBL channels. We calculated the Pearson's correlation coefficient for 40 DMSO-treated and 41 ActD-treated nucleoli.

For in vitro condensates, we calculated the Pearson's correlation coefficient for entire condensates using masks generated using fluorescence intensities of the NPM1, SURF6, and rRNA channels.

RDF analysis

To calculate the RDF of intensities surrounding FC centers in cells, we used the Cellprofiler plugin "MeasureRDF" and associated RDF calculation python code developed by Quinodoz et al.[23] with parameters slightly adjusted to optimize segmentation in our dataset. In brief, as suggested by Quinodoz et al. we utilized Snakemake[48] to impose a consistent workflow, which first projected files to a maximum intensity profile, then supplied images to the MeasureRDF pipeline. We used Cellprofiler[49] to denoise the GC channel, then segmented distinct GC using an adaptive 3-class thresholding method. Within these GC objects, FC objects were segmented using the appropriate channel and a Global Robust Background thresholding method.

FC objects were then used as center points for a RDF calculation, using the python code provided with the Cellprofiler Pipeline. Microscopy data for individual cells were processed as above, with data from multiple cells averaged, and error calculated. This data was then plotted via GraphPad Prism software. From the RDF intensities, pair-wise Pearson correlation coefficient (PCC) were computed for each cell separately using the cor() function in R 4.1.0. Boxplots were made using ggplot2 and p-value from t-test were computed using the ggpubr package in R.

Recombinant protein expression and purification

Wild-type NPM1 (1-294) and its single cysteine mutant (C21T/C275T)[47]; and wild-type SURF6 (1-361) and its single cysteine mutant (C19S); and SSF1 (1-473) each with an N-terminal 6X poly-histidine tag and a TEV recognition site sub-cloned into pET28 plasmids, were used for protein expression (Tables S2 and S3).

Wild-type NPM1 and its NPM1 (C21T/C275T) mutant were expressed and purified as previously described.[47] Briefly, bacterial cultures were grown up to an optical density (OD_{600}) $\sim$ 0.8 at 37 °C, followed by overnight incubation at 18 °C post-induction with 0.4 mM IPTG. Cells were harvested via centrifugation and lysed in 25 mM Tris, pH 7.5, 300 mM NaCl, 5 mM β-mercaptoethanol (BME) by sonication. NPM1 was extracted from the soluble fraction via Ni-NTA affinity chromatography. His tags were cleaved with TEV protease during overnight dialysis at 4 °C in TBS-150 (10 mM Tris, pH 7.5, 150 mM NaCl, 2 mM DTT). The protein was further purified using a C4 HPLC column, and pure NPM1 protein fractions were lyophilized. The lyophilized protein was re-suspended in a buffer consisting of 10 mM Tris, pH 7.5, 2 mM DTT, and 6 M GuHCl and subjected to refolding via dialysis against TBS-150 at 4 °C overnight.

BL21 (DE3) *E. coli* bacteria transformed with plasmids encoding wild-type SURF6 or SURF6-C19S were grown to O.D. $\sim$ 0.8 and induced for protein expression with 0.4 mM IPTG for 4 hours at 37 °C. Bacterial pellets were resuspended in 50 mM Tris, pH 8.0, 300 mM NaCl, 5 mM BME, and 0.1% Triton X-100 containing protease inhibitors (SigmaFast A, Millipore) and sonicated in ice. The pellet fraction was collected and resuspended in 50 mM sodium phosphate, pH 7.5, 300 mM NaCl, and 5 mM BME, 6 M GuHCl and homogenized again by sonication. The resuspension was centrifuged to separate undissolved cell debris, and the supernatant was passed through a Ni-NTA column in the presence of 6 M urea. SURF6-containing fractions were dialyzed at 4 °C in 20 mM Tris, pH 7.5, 300 mM NaCl, and 0.5 mM TCEP in the presence of TEV protease. The cleaved protein was further purified on a Mono-S column using a shallow salt gradient from 150 mM to 1 M NaCl in 20 mM Tris, pH 7.5, 6 M urea, and 5 mM DTT. The protein-containing fraction was dialyzed in 10 mM Tris, pH 7.5, 1 M NaCl, and 2 mM DTT at 4 °C overnight.

BL21 (DE3) *E. coli* bacteria transformed with plasmids encoding SSF1 were grown to O.D. $\sim$ 0.8 and induced for protein expression with 0.4 mM IPTG for 16 hours at 18 °C. The bacterial pellet obtained was processed using the same protocol used for SURF6. Additionally, Ni-NTA chromatography was performed using the same protocol as SURF6. SSF1-containing fractions were dialyzed at 4 °C in 2 M urea, 20 mM Tris, pH 7.5, 300 mM NaCl, and 0.5 mM TCEP in the presence of TEV protease. The cleaved protein was further purified on a Mono-S column using a shallow salt gradient from 300 mM to 1 M NaCl in 20 mM Tris, pH 7.5, 6 M urea, and 5 mM DTT. The protein-containing fractions were further purified using a C4 HPLC column, and pure SSF1 protein fractions were lyophilized. The lyophilized protein was re-suspended in a buffer consisting of 20 mM Tris, pH 7.5, 2 mM DTT, and 6 M GuHCl and subjected to refolding via 10 mM Tris, pH 7.5, 1 M NaCl, and 2 mM DTT at 4 °C overnight.

Expression and purification of 70S ribosomes and total rRNA

The 70S ribosomes (assembled $rRNA_{EC}$) were purified from E. coli K12, strain A19, grown in Luria Broth. Cultures were grown at 37 °C to an $OD_{600} \sim 0.8$ and harvested by centrifugation. Cell pellets were suspended in B-Per reagent (Thermo Fisher Scientific) supplemented with lysozyme and Turbo DNase (Thermo Fisher Scientific). The cell lysate was centrifuged at 30,000 g, 4 °C for 1 h. The soluble fraction was layered onto a 30% sucrose cushion prepared in 20 mM Tris-HCl, pH 7.5, 50 mM MgOAc, 100 mM NH_4Cl, 2 mM DTT, 0.5 mM EDTA buffer (buffer D) and centrifuged at 30,000 g, for 16 h at 4 °C in a SW 32 Ti Swinging-Bucket Rotor (Beckman Coulter). The resulting pellet was gently suspended in buffer D and subjected to two additional rounds of purification on a 30% sucrose cushion. The resulting pellet was dissolved in 10 mM Tris-HCl, pH 7.5, 6 mM MgOAc, 50 mM NH_4Cl, 1 mM TCEP, 0.5 mM EDTA buffer, and centrifuged at 16,000 g, 4 °C for 10 mins. The supernatant containing pure 70S ribosomes was aliquoted and stored at -80 °C. For phase separation assays, 70S ribosomes were dialyzed against TBS-150 containing 6 mM $MgCl_2$ and further stabilized through the addition of 3 mM $MgCl_2$. Total rRNA (unassembled $rRNA_{EC}$) was extracted from the 70S ribosomes using the chloroform:isoamyl alcohol Trizol extraction protocol,[50] and total rRNA purity was estimated by measuring the A260/A280 ratio. Samples with ratios ~2.0 were further analyzed on a 1.2% 1X TBE agarose gel for quality control. Purified total rRNA was stored at -80 °C. The total rRNA stock was diluted into TBS-150 buffer for the phase separation assays.

Preparation of human ribosomal RNA from DLD-1 cells

Human ribosomal RNA was isolated from cytoplasmic ribosomes derived from DLD-1 cells. Cells were harvested and lysed using Cell Lysis Buffer (20 mM Tris-HCl, pH 7.5, 130 mM KCl, 2.35 mM $MgCl_2$, 0.5% Triton X-100, 1 mM DTT, 200 U/ml RNase inhibitor [NEB, M0314], and protease inhibitor cocktail [Thermo Fisher, 78425]). The lysate was centrifuged at 12,000 × g for 5 minutes to separate the cytoplasmic fraction. To dissociate non-ribosomal RNA species (e.g., tRNA and mRNA), the supernatant was supplemented with KCl and puromycin to final concentrations of 300 mM and 1 mM, respectively. The mixture was incubated on ice for 20 minutes, followed by incubation at 37 °C for 15 minutes. Ribosomes were subsequently pelleted by ultracentrifugation through a sucrose cushion (1 M sucrose, 30 mM HEPES, pH 7.5, 100 mM KCl, 5 mM $Mg(OAc)_2$) at 120,000 rpm for 1 hour using a TLA-120.2 rotor. The resulting ribosomal pellet was resuspended in TRIzol reagent, and ribosomal RNA (unassembled $rRNA_{HU}$) was extracted from the aqueous phase and precipitated using isopropanol.

Isolation of pre-60S assembly intermediates from DLD-1 cells

Pre-60S ribosomal particles (assembled $rRNA_{HU}$) were isolated from DLD-1 cells using a modified protocol based on a previously described method.[15] Cells were pre-treated with 100 μg/ml cycloheximide and lysed in Cell Lysis Buffer containing 25 mM HEPES (pH 7.6), 65 mM NaCl, 65 mM KCl, 3 mM $MgCl_2$, 10% glycerol, 0.05% Triton X-100, 1 mM DTT, protease inhibitor cocktail, 20 U/ml TURBO DNase, and 200 U/ml RNase inhibitor. Lysis was performed by bouncing with pestle A (7 strokes) on ice. To assess the sedimentation coefficients of nuclear pre-ribosomal complexes in reference to the established profiles of cytoplasmic ribosomal particles, the clarified cytoplasmic fraction was subjected to sucrose gradient centrifugation (10–45%) using the same conditions described in the "Isolation of 60S Ribosomal Subunits from DLD-1 Cells." Nuclei were washed twice with Cell Lysis Buffer and subsequently resuspended in Nuclear Lysis Buffer containing 25 mM HEPES (pH 7.6), 10 mM KCl, 10 mM $MgCl_2$, 1 mM $CaCl_2$, 1 mM EDTA, 100 mM arginine, 25 mM ATP, 1 mM spermidine, 5% glycerol, 0.1% Triton X-100, 1 mM DTT, protease inhibitor cocktail, and 20 U/ml TURBO DNase. Nuclear lysis was carried out by incubation on a nutator at 4 °C for 30 minutes. Insoluble material was removed by centrifugation at 20,000 × g for 20 minutes at 4 °C. The resulting nuclear supernatant was adjusted to 75 mM KCl and 75 mM NaCl, layered onto a 10–30% sucrose gradient prepared in buffer containing 10 mM Tris-HCl, pH 7.5, 150 mM NaCl, 6 mM $MgCl_2$, and 2 mM DTT, and centrifuged at 36,000 rpm for 3 hours at 4 °C using a Beckman SW41Ti rotor. Further, the pre-60S fractions were pooled and a western blot was performed to probe for pre-60S early assembly factors including DDX18, PPAN (SSF1), and SURF6 (Figure S4G). The pre-60S fractions were dialyzed into 10 mM Tris-HCl, pH 7.5, 150 mM NaCl, 6 mM $MgCl_2$, and 2 mM DTT and concentrated for performing phase separation assays.

Negative stain electron microscopy

Pre-60S ribosomal particles were dialyzed into 10 mM Tris-HCl, pH 7.5, 150 mM NaCl, 6 mM $MgCl_2$, and 2 mM DTT and concentrated to 62.3 ng/μL. The sample was applied to Carbon Film 400 Mesh, CU grids (Electron Microscopy Sciences) which were plasma cleaned for 20 s using a Solarus II Plasma Cleaner (Gatan). 3 μL sample was applied to grid and incubated for 3 minutes and blotted with filter paper. Then, an additional 3 μL sample was applied to grid, incubated for 3 minutes and blotted again. The sample was stained with 3% aqueous uranyl acetate (Figure S4H). Images were acquired on a Thermo Fischer Talos L120C equipped with a Ceta-M camera and a LaB6 electron source operated at 120kV. Images were collected at 57,000x magnification using SerialEM software at 2.55 Å per pixel.

Fluorescence labeling

Fluorescently labeled NPM1 was prepared as previously reported.[47] Briefly, lyophilized NPM1 (C21T/C275T) was resuspended in a 25 mM Tris, pH 7.5, 2 mM TCEP, containing 6 M GuHCl and mixed with a two-fold molar excess of Alexa Fluor 488 maleimide fluorophore (Invitrogen). Unbound dye and unlabeled proteins were removed through HPLC purification. To ensure that each NPM1 pentamer contains only a single fluorophore, labeled and unlabeled NPM1 were refolded at a 1:10 ratio.

SURF6 (C19S) was labeled with Alexa Fluor 647 maleimide (Invitrogen) and SSF1 was labeled with Alexa Fluor 555 NHS ester (Invitrogen) at 1:2 protein:dye ratio under denaturing conditions (10 mM Tris, pH 7.5, 2 mM DTT, 6 M GuHCl) overnight at 4 °C. Samples were dialyzed in TBS-1M at 4 °C. Any remaining unbound dye was removed by ultrafiltration using a 10 kDa cut-off filter (Amicon).

70S ribosomes were labeled with Alexa Fluor 555 NHS ester (Invitrogen) at 1:2 ribosome:dye ratio in 50 mM sodium bicarbonate, pH 8.3, 6 mM $MgCl_2$, 150 mM NaCl, 2 mM DTT overnight at 4 °C. Samples were dialyzed in 10 mM Tris, pH 7.5, 2 mM DTT, 150 mM NaCl, 6 mM $MgCl_2$ at 4 °C. Any remaining unbound dye was removed using a PD-10 column (Cytiva).

Preparation and imaging of ternary and binary condensates

To prepare the NPM1-SURF6-rRNA ternary condensates, we pre-made NPM1-rRNA condensates by mixing 2% AF488-NPM1 and rRNA in TBS-150 containing 2 μM SYTO 40 dye (for rRNA visualization) on a 16-well chambered slide (Grace Bio-Labs) coated with Sigmacote and 1% Pluronic F-127. NPM1-rRNA condensates were allowed to settle at the bottom for 40 minutes at room temperature. Following this, 2% labeled SURF6 was added to the NPM1-rRNA condensates. Condensates were allowed to equilibrate for three hours prior to imaging. Similarly, to prepare the NPM1-SSF1-SURF6-rRNA quaternary condensates, we pre-made NPM1-SSF1-rRNA condensates by mixing 2% AF488-NPM1, 2% AF555-SSF1, and rRNA in TBS-150 containing 2 μM SYTO 40 dye. NPM1-SSF1-rRNA condensates were allowed to settle at the bottom for 40 minutes at room temperature. Following this, 2% labeled SURF6 was added to the NPM1-SSF1-rRNA condensates. For imaging multiphase condensates in the presence of RNase A, 100 ng/μL of RNase A was added into the pre-formed multiphase condensates, and incubated at 37 °C for 30 mins. Images were collected on a Zeiss LSM 780 Observer.Z1 through a Plan Apochromat 63X/1.4 objective lens. All the images were processed using Fiji image processing software,[51] and the data were plotted GraphPad Prism 10 software.

Shell and core phase fusion events and the SURF6 accumulation experiments were recorded on a 3i Marianas system configured with a Yokogawa CSU-W1 confocal scanning system using an alpha Plan Apochromat 100x/1.46 objective. For the SURF6 accumulation experiments, images for the NPM1 channel, in addition to the SURF6 channel, were recorded to demarcate the shell phase. Two-channel images of the condensates were collected every 30 seconds. To measure the increase in SURF6 intensities over time for the two phases, we defined 1-micron diameter ROIs and recorded the mean intensities that were contained within either the core or shell phases. To correct for photobleaching resulting from image acquisition, we measured the rate of photobleaching of multiphase condensates labeled with 2% SURF6. The photobleaching rate was extracted from single exponential fits of the curve .[20]

Phase diagrams of the binary condensates were generated using 2% fluorescently labeled proteins to visualize NPM1 or SURF6 over the range of reported concentrations. For condensates containing rRNA, 2 μM SYTO 40 dye was added. Samples were incubated for two hours prior to imaging. Images were collected on the 3i Marianas system described above and were processed using Fiji image processing software.[51]

ANS fluorescence measurements

10 μM 8-Anilino-1-naphthalenesulfonic acid ammonium salt (ANS) (Sigma-Aldrich, USA) was added to binary condensates. For ternary condensates, ANS dye was mixed with NPM1 and rRNA, followed by a 40-minute incubation period. Subsequently, SURF6 was added to the preformed NPM1-rRNA condensate solution, followed by incubation at room temperature for approximately 2 h before imaging in CultureWell 16-well chambered slides to reach equilibrium conditions. All images were collected on a Zeiss LSM 780 Observer.Z1 through a Plan Apochromat 63X/1.4 objective lens. ANS fluorescence was excited with a 405 nm diode laser, and emitted light was collected at 420-624 nm. All images were processed using Fiji image processing software,[51] and the data were plotted GraphPad Prism 10 software.

Nile red fluorescence measurements

The nile red (Sigma-Aldrich, USA) stock was made in DMSO. 5 μM of nile red was added to pre-formed binary and ternary (multiphase) condensates. Imaging was performed in CultureWell 16-well chambered slides. All images were collected on a Zeiss LSM 780 Observer.Z1 through a Plan Apochromat 63X/1.4 objective lens. Nile red fluorescence was excited with a 561 nm laser, and emitted light was collected at 590-740 nm. All the images were processed using Fiji image processing software[51] and the data were plotted GraphPad Prism 10 software.

Engulfment of cores by shells and contact angle measurements

Chambered slides (Grace BioLabs, Bend, OR) were treated with 1% Pluronic F-127 solution (Sigma-Aldrich, USA), to make the surface hydrophilic. To prepare the core mimics, 25 ng/μL of rRNA and 2.5 μM of SURF6 were mixed, and for the shell, 5 μM NPM1 and 4 μM of SURF6 were mixed. After 40 mins, the core mimics were added to the pre-formed shell mimics and analyzed using time-lapse imaging. Images were acquired on a Zeiss Lattice Light Sheet 7 microscope. Volumetric image stacks were captured using dithered square virtual lattices (Sinc3 30x1000; approximately 30 μm in length and 1,000 nm in width) and stage scanning with 0.2 μm step sizes, resulting in 145 nm deskewed z-steps. Images were captured using a pair of Hammamatsu Orca Fusion sCMOS cameras, with the emission cube comprised of a 640 beam splitter, a 495-550 emission filter for AF488, and a long-pass 655 filter for AF647. Volumes were captured once every 6 seconds using 5 ms planar exposures. Images were saved in .czi file format and subsequently deskewed using the Zeiss Zen Blu2 3.6 Software. Deskewed lattice light sheet volumes were converted into Arivis SIS files using the Arivis 4 Software (Zeiss).3D reconstruction and subsequent images and movies were created and exported from Arivis 4D software. For contact angle measurements, three binary condensates were prepared: NPM1-SURF6 (10 μM each), NPM1-rRNA (10 μM NPM1 with 50 ng/μL rRNA), and SURF6-rRNA (10 μM SURF6 with 50 ng/μL rRNA). Deskewed lattice light sheet volume stacks were acquired and projected in XZ to obtain the shape profile. The contact angles, which are the angle between the line tangent to the condensates and the contact line interior of the condensate, were measured using Fiji image processing software,[51] and the data were plotted GraphPad Prism 10 software.

Determination of volume fraction, K_{app}, and $\Delta\Delta G^{Efflux}$

We analyzed the volume fractions of the cores $Vf_{(core)}$ within multiphase droplets from the volume of the core regions, $V_{(core)}$, and the volume of the entire condensate $V_{(total)}$ (Equation 1). $V_{(core)}$ was determined by creating surface masks from thresholding the mean intensities of the rRNA channel. Masks to determine $V_{(total)}$ were created from thresholding the sum of mean intensities of NPM1, SURF6, and rRNA to ensure coverage of the entire volume of condensates. $Vf_{(core)}$ values were determined for individual condensates.

$$Vf_{(core)} = V_{(core)}/V_{(total)} \qquad \text{(Equation 1)}$$

The apparent partition coefficient (K_{app}) of NPM1 in homogeneous and multiphase condensates as a function of SURF6 concentrations was determined from mean intensities within the segmented masks from micrographs (Equations 2, 3, and 4). For determining the $K_{app\ (dense)}$ of homogeneous condensates, the NPM1 mean intensities inside homogenous condensates, $I_{NPM1\ dense}$, were extracted from masks segmented from the sum of all channel intensities, which covered the entire condensate area. For the multiphase condensates, separate masks that represent the shell and core sub-phases of the condensates were generated. The mean intensities from the NPM1 (enriched component in the shell phases) channel were used to create the masks for the shell regions, while the rRNA (enriched component in the core phases) channel was used for the core regions. Following this, the NPM1 mean intensities ($I_{NPM1,\ shell}$ and $I_{NPM1,\ core}$) were determined within the masks. To account for background signals, intensities from images of buffer solutions collected using identical imaging parameters were subtracted. To measure the light phase NPM1 intensities ($I_{NPM1\ light}$), areas outside the condensates were masked, and the intensity threshold cut-off was manually adjusted when necessary to remove areas with out-of-focus condensates. Using the extracted mean intensities, the apparent partition coefficients, K_{app}, were calculated from the NPM1 mean intensities (background subtracted), as previously reported.[18] All visualization and analyses of volumes and mean intensities were performed using IMARIS software.

$$K_{app}\ (\text{homogenous}) = I_{NPM1\ dense} \Big/ I_{NPM1\ light} \qquad \text{(Equation 2)}$$

$$K_{app}\ (\text{core}) = I_{NPM1,core} \Big/ I_{NPM1\ light} \qquad \text{(Equation 3)}$$

$$K_{app}\ (\text{shell}) = I_{NPM1,shell} \Big/ I_{NPM1\ light} \qquad \text{(Equation 4)}$$

K_{app} values (Equation 5) of unassembled $rRNA_{HU/EC}$, assembled $rRNA_{HU}$ (pre-60S ribosomal particles), and assembled $rRNA_{EC}$ (70S ribosomes) within NPM1-SURF6 condensates were calculated from the mean intensities of SYTO 40 dye from the masks of the entire condensate ($I_{SYTO\ 40\ dense}$) at different ratios of NPM1:SURF6 and the intensities from the light phase mask ($I_{SYTO\ 40\ light}$). Transfer free energy (ΔG^{tr}) values were calculated from the K_{app} values to evaluate the stability of unassembled $rRNA_{HU/EC}$ or assembled $rRNA_{HU/EC}$ (pre-60S/70S) interactions within the condensates (Equation 6). To determine the extent of destabilization of unassembled $rRNA_{HU/EC}$ and assembled $rRNA_{HU/EC}$ (pre-60S/70S) interactions, the change in rRNA transfer free energy was calculated as a function of increased NPM1 concentration, $\Delta\Delta G^{tr}$ (shown in Equation 7). $\Delta G^{tr}_{(2:1)}$ and $\Delta G^{tr}_{(0.7:1/1:1)}$ correspond to transfer free energies at different NPM1:SURF6 ratios, respectively. The change in transfer free energies was transformed to $\Delta\Delta G^{Efflux}$ (Equation 8) to represent the favorability for exclusion of the unassembled $rRNA_{HU/EC}$ and assembled $rRNA_{HU/EC}$ from condensates as interactions were weakened through the addition of excess NPM1.

$$K_{app}\ (\text{rRNA}) = I_{SYTO\ 40\ dense}/I_{SYTO\ 40\ light} \qquad \text{(Equation 5)}$$

$$\Delta G^{tr} = -RT\ln K_{app} \qquad \text{(Equation 6)}$$

$$\Delta\Delta G^{tr} = \Delta G^{tr}_{(2:1)} - \Delta G^{tr}_{(0.7:1/1:1)} \qquad \text{(Equation 7)}$$

$$\Delta\Delta G^{Efflux} = 1 \Big/ \Delta\Delta G^{tr} \qquad \text{(Equation 8)}$$

Bioinformatics analyses of the pre-60S assembly factors

To identify unresolved regions in the cryo-EM structures of assembly factors in Homo sapiens,[15,52] we performed pairwise sequence alignments between the resolved sequences and their corresponding reference sequences from pre-60S assembly states A-H, encompassing a total of 36 assembly factors. Regions that are 30 amino acids long or longer were considered "resolved" for these analyses, and for each assembly factor, the state with the smallest and fewest resolved regions was retained for further analysis. Additionally, the unresolved regions, defined as those outside the resolved regions that contained segments shorter than 30 amino acids, were excluded from the analysis. The retained unresolved regions were analyzed using IUPRED2A[53,54] (https://iupred2a.elte.hu/) to exclude ordered amino acid segments 30 amino acids long or longer with a disorder prediction score below 0.45. Finally, the fractions of arginine (Arg) and lysine (Lys) residues in the unresolved regions were calculated and compared to the human IDRome containing all IDRs in the human proteome. The human IDRome was generated from all human Swiss-Prot (reviewed) proteins contained in Uniprot release 2023_04. From this human proteome dataset, we identified 12,899 unique IDR sequences (from 8,067 proteins) using sak.stjude.org[55] with lengths $\geq$ 60 residues, which is defined as the human IDRome.

QUANTIFICATION AND STATISTICAL ANALYSIS

Detailed descriptions of all quantification methods and experimental parameters are provided in the method details section. Image processing, intensity-based measurements and volumetric analyses were performed using IMARIS software and Fiji. RDF analyses were performed using the CellProfiler plugin "MeasureRDF" with associated Python code. Data were plotted using GraphPad Prism 10 (GraphPad Software), except for Pearson correlation comparisons from RDF-based analyses and the fraction of Arg and Lys residues, which were plotted using the ggplot2 package in R (v4.1.0). The exact values of n and what n represents are provided in the figure legends. Error bars represent SEM or SD as indicated in the figure legends. All box plots show the median (center line), 25th–75th percentiles (box), and minimum–maximum values (whiskers). Statistical comparisons were performed using the two-tailed paired t-test, unpaired two-tailed t-test, Welch's unpaired t-test, or two-tailed Mann-Whitney test, as appropriate for each dataset and as specified in the figure legends. Exact *p*-values and significance indicators are reported in each figure and figure legend. All experiments were performed in three independent biological replicates